

Dell Technologies Enterprise SONiC for PowerFlex

Deployment Guide

Abstract

This guide describes configuring Dell Technologies Enterprise SONiC PowerSwitch systems for single-rack or multirack PowerFlex appliance deployments. BGP EVPN is used in the multirack leaf-spine deployment.

Dell Technologies Solutions

Copyright

Copyright © Dell Inc. All Rights Reserved. Dell Technologies, Dell, and other trademarks are trademarks of Dell Inc. or its subsidiaries. Other trademarks may be trademarks of their respective owners.

Revision history.....	5
Chapter 1: Introduction.....	6
Dell Technologies' vision.....	6
Dell Technologies Enterprise SONiC.....	6
PowerFlex.....	6
Document purpose.....	6
Typographical conventions.....	7
Attachments.....	7
PDF attachments.....	7
HTML attachments.....	7
We value your feedback.....	7
Chapter 2: Hardware overview.....	8
Introduction.....	8
Dell PowerSwitch S5248F-ON.....	8
Dell PowerSwitch S5232F-ON.....	8
Dell PowerSwitch S3248T-ON.....	8
Dell PowerFlex R650 appliances and custom nodes.....	9
Chapter 3: Topology.....	10
Introduction.....	10
Production topology.....	10
BGP EVPN with VXLAN overview.....	12
Production topology connection details.....	12
OOB management topology.....	14
OOB management connection details.....	14
Chapter 4: Deployment planning.....	16
Minimum networking requirements.....	16
Maximum deployment size.....	16
Supported Dell PowerSwitch systems.....	17
VLANs and IP addresses.....	17
LAG to VLAN mapping.....	18
Loopback addresses and BGP ASNs.....	18
Chapter 5: Dell PowerSwitch configuration.....	20
Introduction.....	20
Leaf switch configuration.....	20
General settings.....	21
Configure VLANs.....	22
Configure loopback interfaces.....	23
Configure the MC-LAG domain.....	24
Configure port channels.....	25
Configure management and Ethernet interfaces.....	28

Configure unnumbered BGP for the nondefault VRF.....	31
Configure unnumbered BGP for the default VRF.....	32
Configure VXLAN.....	34
Spine switch configuration.....	35
General settings.....	35
Configure the loopback interface.....	35
Configure management and Ethernet interfaces.....	36
Configure unnumbered BGP.....	37
Verify leaf switch configuration.....	38
show interface status.....	39
show lldp table.....	39
show mclag brief	40
show vlan.....	40
show ip route.....	41
show ip route vrf <i>vrf_name</i>	42
show ip vrf.....	42
show vxlan tunnel.....	43
show vxlan remote vni.....	43
Verify spine switch configuration.....	44
show interface status.....	44
show lldp table.....	44
show ip route.....	45
Chapter 6: Deploy PowerFlex.....	46
Deployment notes.....	46
Appendix A: Using an access-aggregation topology with PowerFlex.....	47
Introduction.....	47
Access-aggregation topology networking requirements.....	48
Topology differences.....	49
Switch configuration file examples.....	49
Appendix B: Validated components.....	50
Introduction.....	50
Dell PowerSwitch systems.....	50
PowerFlex nodes.....	50
PowerFlex Management Controller	51
Appendix C: Resources.....	52
Dell Networking resources.....	52
PowerFlex guides.....	52
Fabric Design Center.....	52
Technical support.....	52

Revision history

Table 1. Document revision history

Part number	Release date	Description of changes
H19678.3	April 2026	Updated mentions of lacp callback to lacp individual. Updated N3248TE-ON to S3248T-ON.
H19678.2	March 2024	Added an appendix: Using an access-aggregation topology with PowerFlex. Additional minor updates related to access- aggregation topology.
H19678.1	January 2024	Minor updates to the following sections: Document purpose, Production topology, OOB management topology, OOB management connection details, and Minimum networking requirements
H19678	August 2023	Initial release.

Introduction

Topics:

- [Dell Technologies' vision](#)
- [Dell Technologies Enterprise SONiC](#)
- [PowerFlex](#)
- [Document purpose](#)
- [Typographical conventions](#)
- [Attachments](#)
- [We value your feedback](#)

Dell Technologies' vision

The vision at Dell Technologies is to be the essential technology company for the data era. Dell Technologies ensures modernization for today's applications and the emerging cloud-native world. Our Networking team is committed to disrupting the market with an open strategy that gives you the freedom of choice for networking operating systems and top-tier merchant silicon. The Dell Technologies strategy enables business transformations that maximize the benefits of collaborative software and standards-based hardware, including lowered costs, flexibility, freedom, and security. Dell Technologies provides further customer enablement through validated deployment guides that demonstrate these benefits while maintaining a high standard of quality, consistency, and support.

Dell Technologies Enterprise SONiC

SONiC is an open-source, Linux-based network operating system (NOS) that runs on switches from multiple vendors and ASICs. Dell Technologies Enterprise SONiC is a hardened and validated version of SONiC supported on selected Dell PowerSwitch platforms. Dell Technologies Enterprise SONiC supports an intuitive command-line interface, object-based administration through a REST interface, and Google's gRPC Network Management Interface (gNMI).

PowerFlex

PowerFlex is a highly scalable compute and storage infrastructure that integrates PowerFlex software-defined storage software with Dell PowerEdge servers. PowerFlex supports a small cluster starting point of four nodes, bring-your-own networking hardware, and bring-your-own management hardware.

PowerFlex allows growth and management of compute and storage resources together or independently depending on business needs. PowerFlex uses PowerFlex Manager for management and operations. PowerFlex Manager enables users to build, automate, and simplify implementation, expansion, and life cycle management.

Document purpose

This guide covers Dell Technologies Enterprise SONiC PowerSwitch configuration for single or multirack PowerFlex deployments. This deployment uses PowerFlex partial network automation (PNA). With PNA, PowerFlex Manager automatically configures the VLANs and IP addresses on the PowerFlex nodes. Dell PowerSwitch network switches running Enterprise SONiC are configured manually.

 **NOTE:** PowerFlex full network automation (FNA) is not supported with Dell Technologies Enterprise SONiC.

Typographical conventions

The CLI and UI examples in this document use the following conventions:

Monospace text	CLI examples
Bold monospace text	• CLI examples which wrap the page • Highlighted information in CLI output
<i>Italic monospace text</i>	Variables in CLI examples
Bold text	• UI fields • Information which is entered in the UI

Attachments

This document includes one or more file attachments. Access the attachments in either of the following ways.

PDF attachments

If you are viewing a .pdf version of this document, access the file attachments as follows:

1. Download the .pdf file to your hard drive, and then open the document in Adobe Acrobat Reader.
2. In the left pane, click the paperclip icon  to see the attachments.

 **NOTE:** If the left pane is not visible, click the arrow icon  on the left side of the page, and then click paperclip icon .

HTML attachments

If you are viewing this content in HTML format, see the last section for links to the file attachments.

We value your feedback

Dell Technologies and the authors of this document welcome your feedback on the solution and the solution documentation. Contact the Dell Technologies Solutions team by [email](#).

Authors: Dell Technologies Network Solutions Engineering, Technical Marketing, and Information Design & Development teams

 **NOTE:** For links to additional documentation for this solution, see the [Dell Technologies Info Hub for Networking Solutions](#).

This document may contain language from third-party content that is not under Dell's control and is not consistent with Dell's current guidelines for Dell's own content. When such third-party content is updated by the relevant third parties, this document will be revised accordingly.

Hardware overview

Topics:

- [Introduction](#)
- [Dell PowerSwitch S5248F-ON](#)
- [Dell PowerSwitch S5232F-ON](#)
- [Dell PowerSwitch S3248T-ON](#)
- [Dell PowerFlex R650 appliances and custom nodes](#)

Introduction

This section briefly describes the hardware that was used to validate the deployment examples in this guide. [Validated components](#) contains a complete listing of hardware and software that is validated for this guide. See [Supported Dell PowerSwitch systems](#) for a list of switches supported for this deployment.

Dell PowerSwitch S5248F-ON

The Dell PowerSwitch S5248F-ON is a 1-Rack Unit (1U), multilayer switch with 48 x 25 GbE, 4 x 100 GbE, and 2 x 200 GbE ports. This guide uses two S5248F-ON switches in each rack as leaf switches.



Figure 1. Dell PowerSwitch S5248F-ON

Dell PowerSwitch S5232F-ON

The Dell PowerSwitch S5232F-ON is a 1U, multilayer switch with 32 x 100 GbE QSFP28 ports and 2 x 10 GbE SFP+ ports. This guide uses four S5232F-ON switches as spine switches.



Figure 2. Dell PowerSwitch S5232F-ON

Dell PowerSwitch S3248T-ON

The Dell PowerSwitch S3248T-ON is a 1U switch with 48 x 1 GbE BASE-T ports, 4 x 10 GbE SFP+ ports, and 2 x 100 GbE QSFP28 ports. This guide uses one S3248T-ON switch in each rack for out-of-band (OOB) management traffic.



Figure 3. Dell PowerSwitch S3248T-ON

NOTE: Other Dell and third-party switches with 1 GbE BASE-T ports can also be used for OOB management traffic.

Dell PowerFlex R650 appliances and custom nodes

PowerFlex R650 appliances and PowerFlex R650 custom nodes are built on the current generation of Dell PowerEdge servers. The deployment example in this guide uses Dell PowerFlex R650 S (Storage) and PowerFlex R650 C (Compute) nodes. Each PowerFlex node in this deployment has 2 x 25 GbE dual-port SFP28 adapters, an embedded 1 GbE BASE-T dual-port NIC, and an iDRAC.



Figure 4. Dell PowerFlex R650 node

NOTE: PowerFlex appliances and PowerFlex custom nodes are supported for this deployment. PowerFlex R640 nodes are also supported, provided they have two 25 GbE dual-port SFP28 adapters. PowerFlex R640 nodes used as PowerFlex management controllers must also have a 1 GbE BASE-T NIC available in addition to the iDRAC. PowerFlex compute nodes with three 25 GbE dual-port SFP28 adapters are also supported.

Topology

Topics:

- [Introduction](#)
- [Production topology](#)
- [BGP EVPN with VXLAN overview](#)
- [Production topology connection details](#)
- [OOB management topology](#)
- [OOB management connection details](#)

Introduction

The topology for this deployment is divided into two major parts:

1. Production, or in-band
2. Out-of-band (OOB) Management

The production topology contains redundant components and is used for all mission-critical and end-user network traffic. The OOB management network is an isolated network for remote management.

Production topology

The production topology that is used in this guide is shown in the figure below.

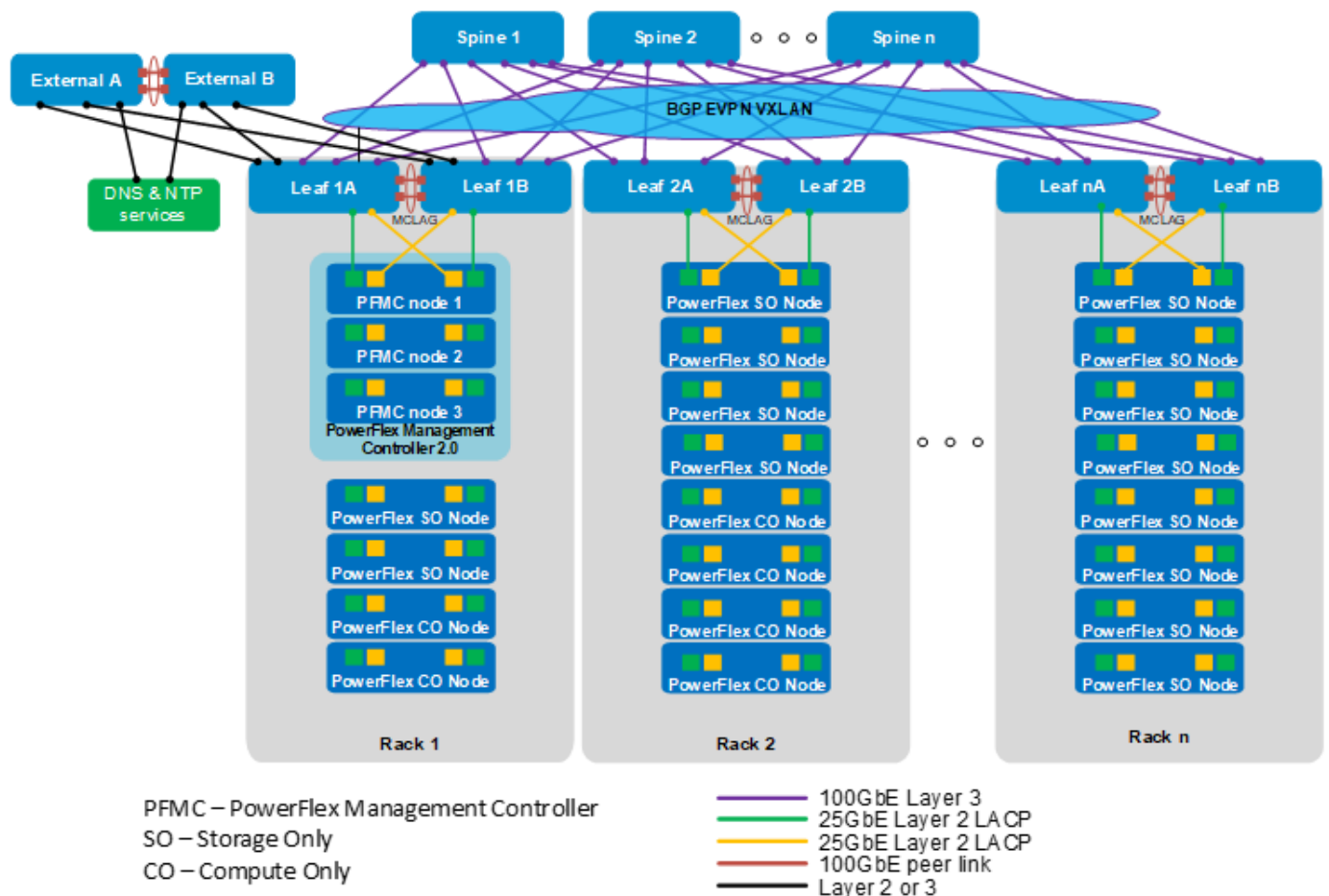


Figure 5. PowerFlex leaf-spine production topology

NOTE: PowerFlex on SONiC is supported with a leaf-spine BGP EVPN topology as shown in this guide. For a leaf and spine deployment, PowerFlex recommends a minimum of 3 spines. In this document, 4 spines are used. An access, aggregation, core topology may be supported in some cases. See Appendix A for more information.

Each rack has two leaf switches connected to each other with a multichassis link aggregation group (MC-LAG) peer link. Every leaf switch is connected to every spine switch to form a leaf-spine (also known as a Clos) topology.

NOTE: Leaf switches are sometimes called top-of-rack (ToR) switches.

In this deployment example, a three-node PowerFlex Management Controller (PFMC) 2.0 cluster is in Rack 1. The remaining nodes in Racks 1-*n* are compute only (CO) and storage only (SO) nodes. This deployment uses four 25 GbE network interface controller ports per PowerFlex node for production network connections, as shown in the figure above.

NOTE: The topology used to validate this guide consisted of four racks, eight leaf switches, four spine switches, and two external switches.

In this deployment example, Leaf1A and Leaf1B are border leaves which are connected to the external switches. The external switches are typically part of an existing network in the data center with DNS and NTP services and Internet or WAN access.

As an option, an additional leaf pair can be added as dedicated border leaves, with no compute or storage nodes connected to them. This option may be beneficial if there is a large amount of North-South traffic between the leaf-spine fabric and the external network. The dedicated border leaf pair must be configured as MC-LAG peers and connected to every spine switch like the other leaf switches in the fabric.

Uplinks from the border leaf switches to the external switches may be configured as Layer 2 or Layer 3. If Layer 3 uplinks are used, you may use Border Gateway Protocol (BGP), Open Shortest Path First (OSPF), or static routing. In the configuration example used in this guide, the uplinks are configured as Layer 3 with BGP.

NOTE: If an external network is not used, the required DNS and NTP services may be on a server that is connected directly to a leaf switch.

Equal-Cost Multi-Path routing (ECMP) is enabled on the switches to use all available bandwidth on the leaf-spine connections. Border Gateway Protocol (BGP) Ethernet VPN (EVPN) with Virtual Extensible LAN (VXLAN) is used to stretch Layer 2 networks across the Layer 3 leaf-spine fabric.

BGP EVPN with VXLAN overview

NOTE: If spines are not used in a single rack deployment, then BGP EVPN with VXLAN is not used.

For multirack deployments, BGP EVPN with VXLAN stretches virtual (overlay) Layer 2 networks across a physical (underlay) Layer 3 leaf-spine fabric. This architecture allows for the scalability of Layer 3 networks with the mobility benefits of a Layer 2 network. For example, a VM or physical host can be moved from one rack to another without changing its IP address and gateway information.

EVPN uses BGP to exchange endpoint MAC and IP address information between VXLAN tunnel endpoints (VTEPs). In this topology, each pair of leaf switches in a rack is one logical VTEP. Packets destined to a remote device are sent to the local VTEP as the first hop. The packet is encapsulated with a VXLAN header that includes the Virtual Network Identifier (VNI) and the IP address of the destination host. See the figure below.

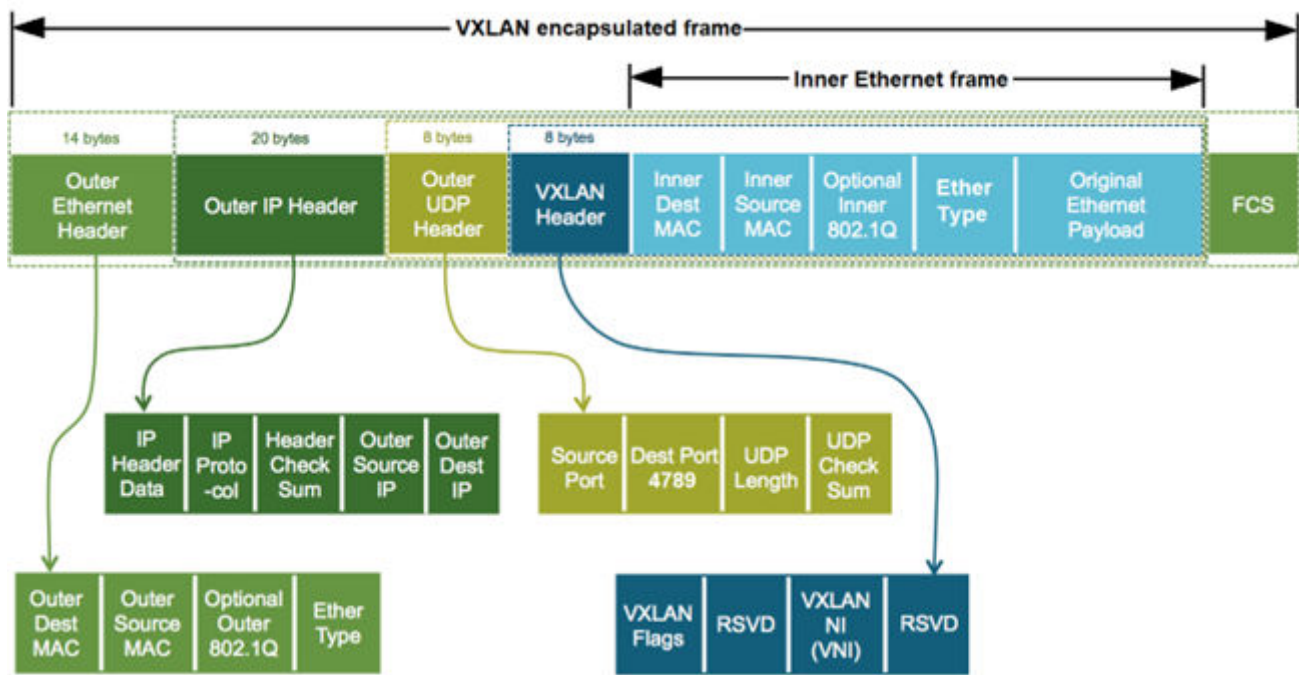


Figure 6. VXLAN encapsulated frame

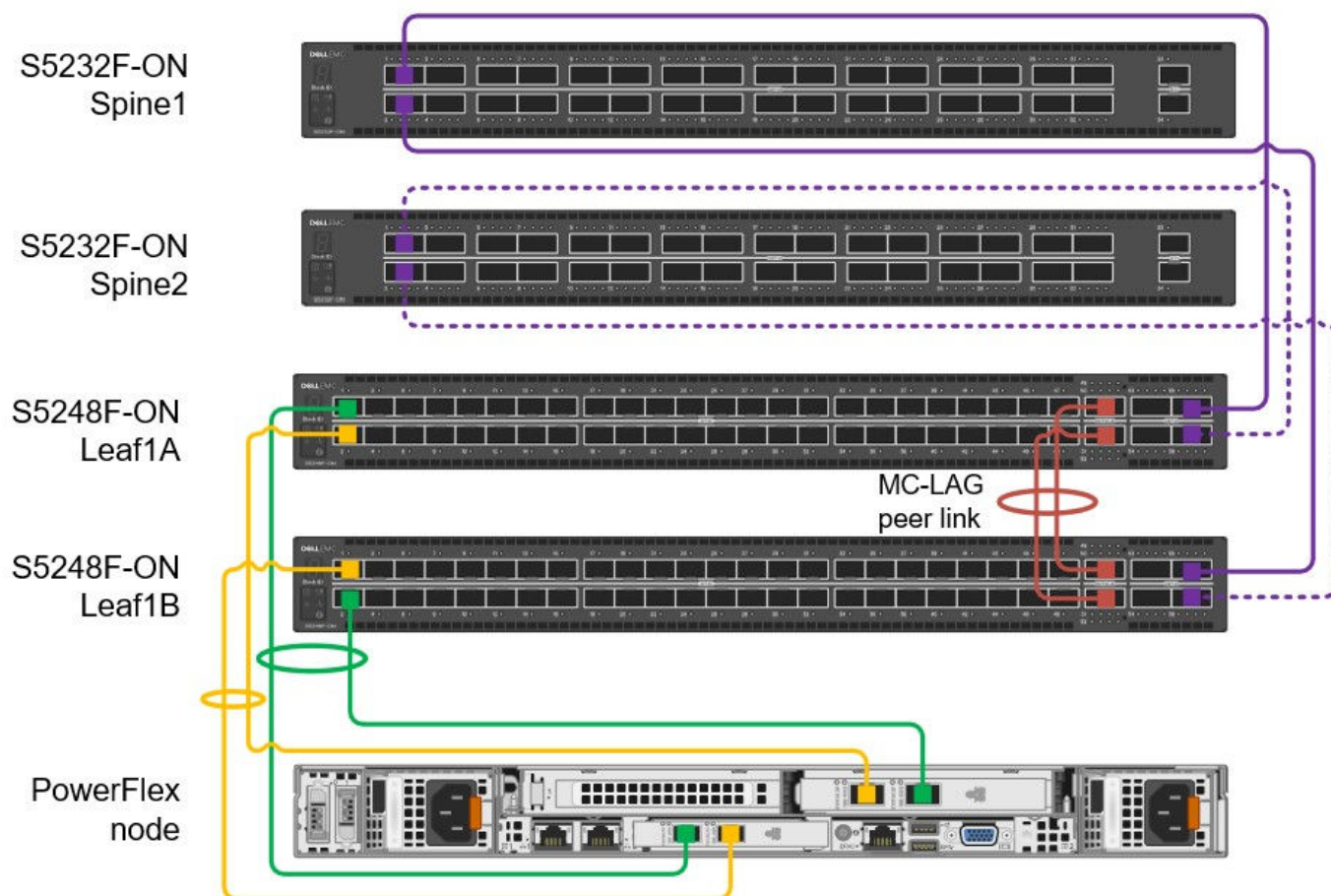
The destination VTEP extracts the original packet and routes it to the final destination. VXLAN encapsulation adds approximately 50 bytes of overhead to each Ethernet frame. As a result, all switches in the underlay (physical) network must be configured to support an MTU of at least 1600 bytes on all participating interfaces.

NOTE: In this deployment example, switch interfaces are set to their maximum supported MTU size of 9216 bytes for best performance.

Anycast gateways with the same IP address are configured on each leaf pair. The anycast gateway IP address is set as the default gateway for all hosts and VMs on that virtual network. Hosts and VMs with anycast gateways use the same gateway information while behind different leaf pairs. This configuration replaces Virtual Router Redundancy Protocol (VRRP) and enables hosts and VMs to migrate from one leaf pair to another without changing the network configuration.

Production topology connection details

Production network connection details for this deployment example are shown in the figure below.



Note: PowerFlex node NIC locations may vary depending on model.

- 25GbE LACP link (FE, Bond0)
- 25GbE LACP link (BE, Bond1)
- 100GbE leaf-spine link (Spine 1)
- - - 100GbE leaf-spine link (Spine 2)
- 200GbE MC-LAG peer link

Figure 7. Production network connection details

NOTE: Other PowerFlex nodes, leaf switches, and spine switches that are not shown are connected in the same manner.

Each leaf switch has one connection to each spine switch. For this deployment, all leaf-spine connections are 100 GbE.

All PowerFlex nodes have two connections to each leaf. The connections in this example are 25 GbE. The four 25GbE connections from each PowerFlex node to the leaf switches are configured in two LACP link aggregation groups (LAGs).

For the MC-LAG peer link connections, two QSFP28-DD double density ports (2 x 100 GbE interfaces per physical port), are available on S5248F-ON switches. They are used to create a 400 GbE port channel that is the MC-LAG peer link. This connection requires QSFP28-DD DAC cables or optics. On switches without QSFP28-DD ports, QSFP28 100 GbE or SFP28 25 GbE ports are used for peer link connections. Dell Technologies recommends using at least two physical ports on each leaf switch for the peer link. This configuration is for redundancy and provides additional bandwidth if there is a failure. The peer link is used for data traffic only if there is a link failure that requires the peer link to reach the destination.

OOB management topology

The out-of-band (OOB) management network is an isolated network for remote management of hardware. This hardware includes PowerFlex nodes and PowerSwitch network switches. It can also include other devices using their dedicated management ports.

One S3248T-ON switch is typically installed in each rack for OOB management network connections, as shown below.

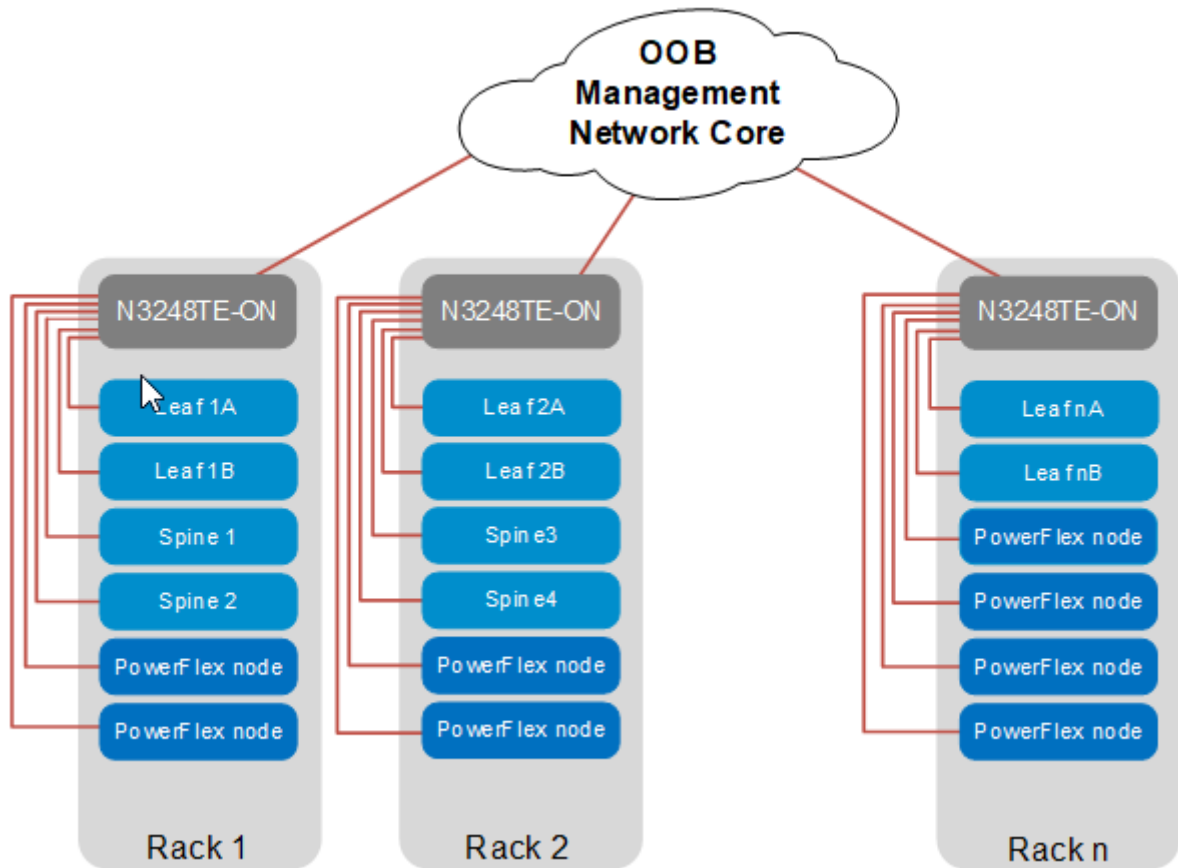


Figure 8. OOB management network topology

The 1 GbE 1000BASE-T ports on each S3248T-ON are connected downstream to hardware management ports on each device in the rack. Those ports include the PowerSwitch management ports and PowerFlex node iDRAC ports. Dedicated management ports on other devices in the rack can also be connected.

Four 10 GbE SFP+ ports are available on each S3248T-ON for use as uplinks to the OOB management network core and for downlinks to the 10 GbE OOB management NIC on the PFMC nodes.

OOB management switch configuration is not detailed in this guide. The OOB management network can be Layer 2 or Layer 3, and can be configured to suit your environment.

NOTE: For reference, devices on the OOB Management network in this guide use the 100.67.0.0/16 IP address range. All addresses that are used in this guide are examples only.

OOB management connection details

All PowerSwitch management ports and PowerFlex node iDRACs are connected to the OOB management switch in each rack.

Each PFMC node has two connections to the OOB management network: the 1 GbE iDRAC and an additional 10 GbE NIC, as shown in the figure below. PowerFlex Manager uses the PFMC node's 10 GbE to communicate with the iDRACs on the PowerFlex compute and storage nodes. The PowerFlex compute and storage nodes only require the iDRAC connection to the OOB network.

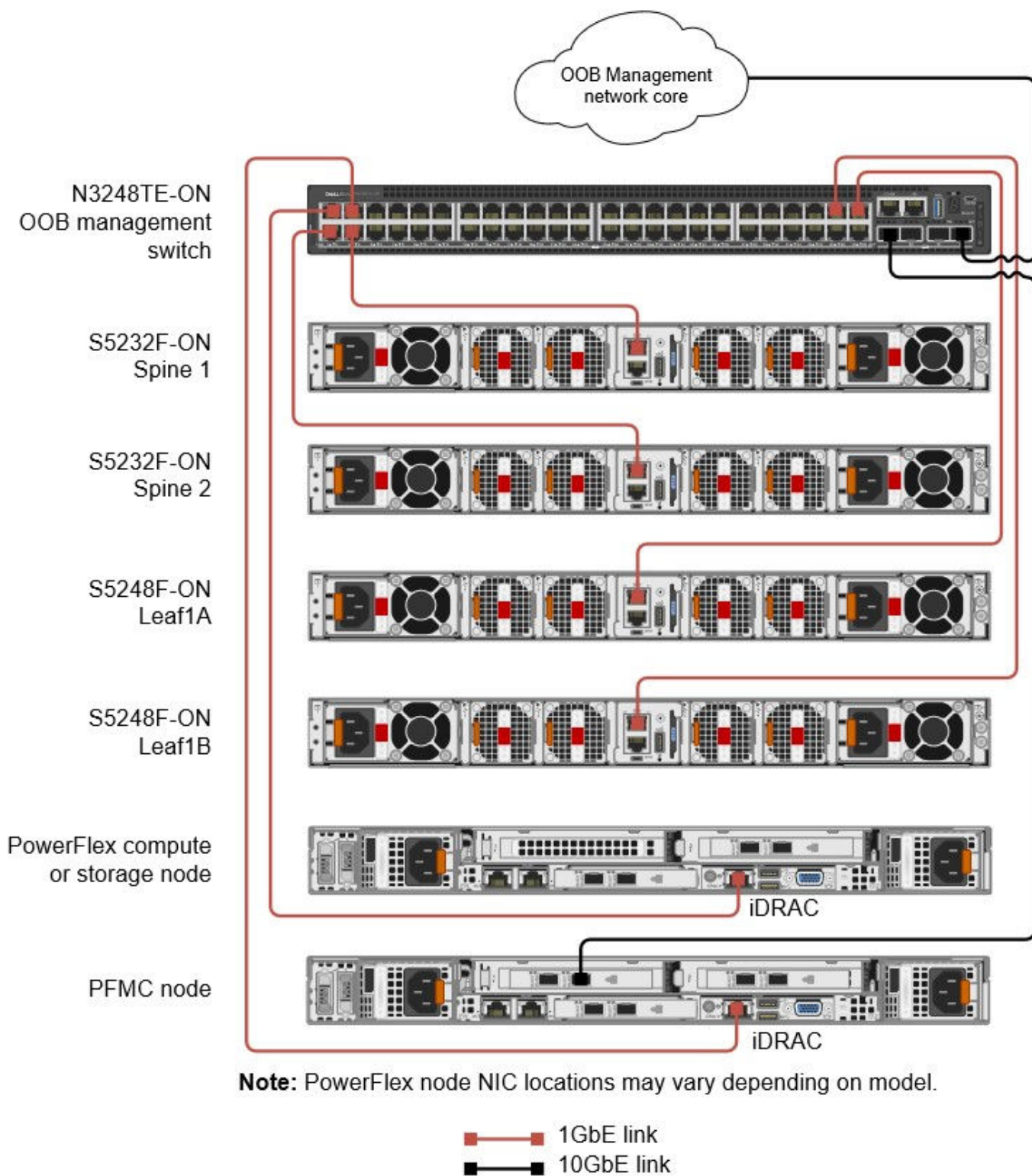


Figure 9. OOB management network connection details

NOTE: Other PowerFlex nodes, spines, and leaf switches that are not shown are connected in the same manner.

Deployment planning

Topics:

- [Minimum networking requirements](#)
- [Maximum deployment size](#)
- [Supported Dell PowerSwitch systems](#)
- [VLANs and IP addresses](#)
- [LAG to VLAN mapping](#)
- [Loopback addresses and BGP ASNs](#)

Minimum networking requirements

NOTE: In this guide, "single rack" means topologies with only two leaf switches. "Multirack" means topologies with four or more leaf switches.

The minimum networking requirements for a PowerFlex on Dell Technologies Enterprise SONiC deployment include the following:

- Single rack deployments require two supported Dell PowerSwitch leaf switches of the same model.

NOTE: If a future expansion from a single to a multirack topology is likely, Dell Technologies recommends installing and configuring two supported Dell PowerSwitch spine switches. Doing so will simplify the expansion process in the future.

- Multirack deployments require two supported Dell PowerSwitch leaf switches of the same model per rack, plus at least two supported Dell PowerSwitch spine switches.
- All Dell PowerSwitch leaf and spine systems in fabric must be running the same version of Dell Technologies Enterprise SONiC.
- Dell Technologies Enterprise SONiC 4.1.1 or later is required. An Enterprise Standard or Enterprise Premium bundle is required for this deployment.
- One 1 GbE BASE-T switch for OOB management connections is required. Dell Technologies recommends using one PowerSwitch S3248T-ON per rack.
- One DNS server and one NTP server that are reachable from the PowerFlex Node Management network are required.
- Each PowerFlex node requires four 25 GbE SFP28 connections to the leaf switches.
- Each leaf to spine connection requires at least one 100 GbE QSFP28 connection. Provide enough connections to ensure the oversubscription ratio at the leaf switches does not exceed 2:1.
- MC-LAG peer links between leaf switch pairs require at least two connections for fault tolerance. Each link can be 25 GbE SFP28, 100 GbE QSFP28, or 200 GbE QSFP28-DD (available on S5248F-ON). If 25 GbE ports are used, Dell recommends using at least four connections for sufficient bandwidth if a switch is taken offline.
- One 1 GbE (1000BASE-T) connection is required for each switch OOB management port.
- One 1 GbE (1000BASE-T) connection is required for each PowerFlex node iDRAC port.
- One 10 GbE connection is required for each PFMC node to access the OOB management network in addition to the PFMC node's 1 GbE iDRAC port.

NOTE: See your PowerFlex documentation for minimum PowerFlex node requirements.

Maximum deployment size

The maximum number of switches and nodes that are supported for PowerFlex on SONiC deployments are:

- Spine switches—6
- Leaf switches—32
- PowerFlex nodes—352 (22 nodes per leaf pair)

Supported Dell PowerSwitch systems

Dell PowerSwitch systems that are supported as leaf or spine switches for PowerFlex on Dell Technologies Enterprise SONiC deployments are shown in the table below.

NOTE: The number of 100 GbE QSFP28 ports available on the leaf switches limits the maximum number of spine switches in this deployment.

Table 2. Supported switches and roles

Switch	Ports	Roles
S5212F-ON	12x 25 GbE SFP28 3x 100 GbE QSFP28	Leaf
S5224F-ON	24x 25 GbE SFP28 4x 100 GbE QSFP28	Leaf
S5248F-ON	48x 25 GbE SFP28 2x 200 GbE QSFP28-DD (See note below.) 4x 100 GbE QSFP28	Leaf
S5232F-ON	32x 100 GbE QSFP28 2x 10 GbE SFP+	Spine

NOTE: If needed, QSFP28-DD ports can be used in 100 GbE mode with QSFP28 connectors for additional leaf-spine connections.

For OOB management, Dell Technologies recommends one PowerSwitch S3248T-ON per rack. Other Dell and third-party switches with 1 GbE BASE-T ports can also be used for OOB management traffic.

VLANs and IP addresses

In this deployment, traffic is divided into different networks as shown in the *Dell PowerFlex Appliance with PowerFlex 4.x Network Planning Guide*. Depending on the traffic type, some networks are Layer 2 (no gateway) and some are Layer 3 (with gateway). The management and site replication networks must be accessible from devices on other networks, so they are Layer 3. The remaining networks are Layer 2.

NOTE: The VLAN IDs and IP addresses used in this guide are examples only, and can be changed as needed.

Table 3. VLAN IDs and network addresses

VLAN ID	Description	Network name	Network	Gateway
103	vCenter high availability	flex-vcsa-ha-L2	192.168.103.0/24	None
105	Node management	flex-node-mgmt-L3	192.168.105.0/24	192.168.105.254
106	Compute vMotion	flex-vmotion-L2	192.168.106.0/24	None
140	PFMC management	pfmc-sds-mgmt-L3	192.168.140.0/24	192.168.140.254
141	PFMC data 1	pfmc-sds-data1-L2	192.168.141.0/24	None
142	PFMC data 2	pfmc-sds-data2-L2	192.168.142.0/24	None
143	PFMC vMotion	pfmc-vmotion-L2	192.168.143.0/24	None
150	Storage management	flex-stor-mgmt-L3	192.168.150.0/24	192.168.150.254
151	Storage data 1	flex-data1-L2	192.168.151.0/24	None
152	Storage data 2	flex-data2-L2	192.168.152.0/24	None

Table 3. VLAN IDs and network addresses (continued)

VLAN ID	Description	Network name	Network	Gateway
161	Site replication traffic 1	flex-repl1-L3	192.168.161.0/24	192.168.161.254
162	Site replication traffic 2	flex-repl2-L3	192.168.162.0/24	192.168.162.254

LAG to VLAN mapping

Each PowerFlex node is a member of several different VLANs depending on its type. VLAN membership by node type is shown in the table below.

Table 4. VLAN membership by node type

PowerFlex node type	VLAN membership
PowerFlex Management Controller (PFMC)	103, 105, 140-143, 150
Compute Only (CO)	105-106, 151-152
Storage Only (SO)	150-152, 161-162

The four network connections from each PowerFlex node to the two leaf switches are configured in two LACP LAGs as shown in [Production network connection details](#). On PFMC nodes running ESXi, the LAGs are named Front End (FE) and Back End (BE). On PowerFlex CO or SO nodes running the intelligent catalog embedded Linux operating system, the LAGs are named Bond0 and Bond1.

The VLANs are distributed between the two LAGs as shown in the table below.

Table 5. VLAN LAG assignments by node type

PowerFlex node type	LAG 1: FE or Bond0	LAG 2: BE or Bond1
PFMC	105, 140, 150	103, 141-143
CO	105-106	151-152
SO	150-151, 161	152, 162

Loopback addresses and BGP ASNs

Loopback addresses are configured on leaf and spine switches for use as router IDs and VTEP addresses. Autonomous System Numbers (ASNs) are configured on the switches for use with Border Gateway Protocol (BGP).

Each leaf or spine switch must have a unique router ID. Each leaf pair shares a VTEP IP address, and an ASN. Spines do not have VTEP addresses, and all spines in the fabric share the same ASN. The external switches are also configured for BGP in this example. They each have unique router IDs and share an ASN that is different from the leaf and spine ASNs.

 **NOTE:** As a best practice, only use IP addresses and ASNs in the private ranges.

The values that are used in this deployment example are listed in the table below.

Table 6. Loopback addresses and BGP ASNs

Switch	Router ID (Loopback 0)	VTEP address (Loopback 1)	ASN
Leaf 1A	10.1.1.1/32	10.0.0.1/32	65101
Leaf 1B	10.1.1.2/32	10.0.0.1/32	65101
Leaf 2A	10.1.1.3/32	10.0.0.2/32	65102
Leaf 2B	10.1.1.4/32	10.0.0.2/32	65102
Leaf 3A	10.1.1.5/32	10.0.0.3/32	65103

Table 6. Loopback addresses and BGP ASNs (continued)

Switch	Router ID (Loopback 0)	VTEP address (Loopback 1)	ASN
Leaf 3B	10.1.1.6/32	10.0.0.3/32	65103
Leaf 4A	10.1.1.7/32	10.0.0.4/32	65104
Leaf 4B	10.1.1.8/32	10.0.0.4/32	65104
Spine 1	10.2.2.1/32	N/A	65200
Spine 2	10.2.2.2/32	N/A	65200
Spine 3	10.2.2.3/32	N/A	65200
Spine 4	10.2.2.4/32	N/A	65200
External A	10.3.0.1/32	N/A	65300
External B	10.3.0.2/32	N/A	65300

Dell PowerSwitch configuration

Topics:

- [Introduction](#)
- [Leaf switch configuration](#)
- [Spine switch configuration](#)
- [Verify leaf switch configuration](#)
- [Verify spine switch configuration](#)

Introduction

This chapter shows the Dell Technologies Enterprise SONiC configuration commands for Leaf1A and Leaf1B for single or multirack deployments. It also shows the commands for Spine1 and Spine2 for multirack deployments.

Configurations for the remaining leaf and spine switches in the example multirack topology are similar, but are not detailed in this chapter. Full configuration files for all switches in this deployment example are provided as attachments to this guide. The files include the SONiC 4.1.1 default settings plus the nondefault settings that are covered in this chapter. To access the files, see [Attachments](#).

NOTE: In the examples in this guide, LACP is used on all leaf switches for PowerFlex node-to-leaf-switch connections. Dell Technologies recommends using LACP in this deployment as shown, but it is not required. Configuration files without LACP for Leaf1A and Leaf1B are attached as examples. These two file names begin with "no_lacp". All other leaf switch configuration file attachments use LACP. The spine switch configurations for this deployment are the same regardless of the LACP configuration on the leaf switches.

The configuration commands that are listed in this chapter can be edited and applied to the applicable switches in the order shown. All switches in this example start at their factory default configuration. A Dell PowerSwitch network switch that is running Dell Technologies Enterprise SONiC can be returned to its factory settings by:

1. Running `write erase install`
2. Answering **y** when prompted
3. Rebooting the switch
4. Log in as `admin`.
5. Enter the default password `YourPaSsWoRd`, and change the password when prompted.

Leaf switch configuration

This section details the configuration of Leaf1A and Leaf1B for single or multirack deployments. Steps that are not required for single rack deployments without spines are noted. There is no harm in applying the additional multirack configuration steps to a single rack environment without spines. If you are installing spine switches in a single rack deployment, be sure to follow all configuration steps.

For multirack deployments, configuration details for the remaining leaf switches in the example topology are similar, but are not shown in this chapter. Full configuration files for all switches in this deployment example are provided as attachments to this guide.

General settings

About this task

Perform these steps to configure general settings on Leaf1A and Leaf1B:

Steps

1. Enter the SONiC Management Framework CLI.
2. Enter configuration mode.
3. Specify a hostname.
4. Change the interface naming mode to standard.



NOTE: Dell Technologies recommends using standard interface naming mode. In default naming mode, port numbers start at 0. In standard naming mode, port numbers start at 1, which matches the numbers that are printed on the switch.
5. Exit back to the Linux shell to activate the changes.

Leaf1A	Leaf1B
<pre>sonic-cli configure terminal hostname Leaf1A interface-naming standard end exit</pre>	<pre>sonic-cli configure terminal hostname Leaf1B interface-naming standard end exit</pre>

6. Enter the SONiC Management Framework CLI.
7. Enter configuration mode.
8. Configure a unique IP anycast MAC address.

This address must be the same on all leaf switches in the leaf-spine fabric.
9. Enable IPv4 and IPv6 anycast addressing.
10. Set the switch spanning tree mode to rapid per-VLAN spanning tree (RPVST).



NOTE: This topology is loop-free and no ports should be blocked by spanning tree when properly cabled and configured. The use of spanning tree is a best practice and is used here as a precaution only.
11. Configure Bridge Protocol Data Filtering Unit (BPDU) filtering for edge ports globally.
12. Create the management virtual routing and forwarding (VRF) instance as a best practice.

This instance provides an isolated routing table for the management interface. The management VRF must be named **mgmt** as shown.
13. Create a nondefault (or tenant) VRF instance.

The VRF name must start with **vrf** and is a text string with up to 15 characters.



NOTE: A nondefault VRF is required to use symmetric integrated bridging and routing (IRB) with VXLAN. In symmetric IRB, both the ingress and egress VTEPs route packets to their final destination.
14. If you created a management VRF, put the SSH server of the switch in the management VRF.

Doing so enables you to access the switch console on the management interface by using SSH.
15. Enable link state tracking for downstream links.

This setting helps to minimize switch failover time if a switch goes offline.

Leaf1A	Leaf1B
<pre>sonic-cli configure terminal</pre>	<pre>sonic-cli configure terminal</pre>

Leaf1A	Leaf1B
<pre> ip anycast-mac-address 00:00:00:11:11:11 ip anycast-address enable ipv6 anycast-address enable spanning-tree mode rapid-pvst spanning-tree edge-port bpdufilter default ip vrf mgmt ip vrf VrfTenant1 ssh-server vrf mgmt link state track mclag timeout 180 downstream all-mclag </pre>	<pre> ip anycast-mac-address 00:00:00:11:11:11 ip anycast-address enable ipv6 anycast-address enable spanning-tree mode rapid-pvst spanning-tree edge-port bpdufilter default ip vrf mgmt ip vrf VrfTenant1 ssh-server vrf mgmt link state track mclag timeout 180 downstream all-mclag </pre>

Configure VLANs

About this task

For each VLAN, perform the following steps:

Steps

1. Create the VLAN and provide a description.
2. Set the maximum transmission unit (MTU) size to 9216 bytes for best performance.
3. On Layer 3 VLANs, enable neighbor suppression to minimize the flooding of ARP broadcast traffic.
4. Bind the VLAN to the tenant VRF.
5. If the VLAN is Layer 3, assign an IP anycast address that is the same on all leaf switches in the leaf-spine fabric.

Leaf1A	Leaf1B
<pre> interface Vlan103 description flex-vcasa-ha-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan105 description flex-node-mgmt-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.105.254/24 ! interface Vlan106 description flex-vmotion-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan140 description pfmc-sds-mgmt-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.140.254/24 ! interface Vlan141 description pfmc-sds-data1-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! </pre>	<pre> interface Vlan103 description flex-vcasa-ha-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan105 description flex-node-mgmt-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.105.254/24 ! interface Vlan106 description flex-vmotion-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan140 description pfmc-sds-mgmt-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.140.254/24 ! interface Vlan141 description pfmc-sds-data1-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! </pre>

Leaf1A	Leaf1B
<pre> interface Vlan142 description pfmc-sds-data2-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan143 description pfmc-vmotion-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan150 description flex-stor-mgmt-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.150.254/24 ! interface Vlan151 description flex-data1-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan152 description flex-data2-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan161 description flex-rep1-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.161.254/24 ! interface Vlan162 description flex-rep2-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.162.254/24 </pre>	<pre> interface Vlan142 description pfmc-sds-data2-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan143 description pfmc-vmotion-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan150 description flex-stor-mgmt-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.150.254/24 ! interface Vlan151 description flex-data1-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan152 description flex-data2-L2 mtu 9216 ip vrf forwarding VrfTenant1 ! interface Vlan161 description flex-rep1-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.161.254/24 ! interface Vlan162 description flex-rep2-L3 neigh-suppress mtu 9216 ip vrf forwarding VrfTenant1 ip anycast-address 192.168.162.254/24 </pre>

 **NOTE:** If spines are not used, then Steps 6 through 8 are not required for a single rack deployment.

6. Create a VLAN for VRF VNI mapping.
7. Provide a description.
8. Bind the VLAN to the nondefault (or tenant) VRF.

Leaf1A	Leaf1B
<pre> interface Vlan2001 description ForVrfVniMapping ip vrf forwarding VrfTenant1 </pre>	<pre> interface Vlan2001 description ForVrfVniMapping ip vrf forwarding VrfTenant1 </pre>

Configure loopback interfaces

About this task

Perform the following steps to configure loopback interfaces on Leaf1A and Leaf1B:

Steps

1. Configure a loopback interface for use as the router ID.
2. Provide a description.
3. Specify a unique IP address for each switch.
4. For multitrack deployments, configure a loopback interface for use as the VTEP address.
5. Provide a description.
6. Specify an IP address that is unique to this leaf pair.
Use the same address on both leaf switches.

Leaf1A	Leaf1B
<pre>interface Loopback 0 description Router-ID ip address 10.1.1.1/32 ! interface Loopback 1 description vtep ip address 10.0.0.1/32</pre>	<pre>interface Loopback 0 description Router-ID ip address 10.1.1.2/32 ! interface Loopback 1 description vtep ip address 10.0.0.1/32</pre>

Configure the MC-LAG domain

About this task

Perform the following steps to configure the MC-LAG domain:

Steps

1. Create an LACP port channel for use with the MC-LAG peer link.
 **NOTE:** Port channels created in Dell Technologies Enterprise SONiC use LACP by default.
2. Provide a description.
3. Assign all VLANs on the switch to the MC-LAG port channel.
For a multitrack deployment, include the VRF VLAN (2001 in this example).
4. Enable the LACP `fast_rate` option to minimize the time required for failure detection.
5. Set the MTU size to 9216 bytes for best performance.
6. Enable the interface.
7. Create the MC-LAG domain with the `mclag domain domain-id` command.
 **NOTE:** Valid MC-LAG domain numbers are 1 to 4095. Only one domain is supported on a switch.
8. For the source IP, specify the router ID of the local switch.
9. For the peer IP, specify the router ID of the peer leaf switch.
10. Specify the port channel to be used as the MC-LAG peer link.
11. Assign an MC-LAG system MAC that is unique to this leaf pair.
Use the same MAC on both leaf switches.

Leaf1A	Leaf1B
<pre>interface PortChannel 100 description "MC-LAG peer link" switchport trunk allowed Vlan 103-106,140-143,150-152,161-162,2001 fast_rate mtu 9216</pre>	<pre>interface PortChannel 100 description "MC-LAG peer link" switchport trunk allowed Vlan 103-106,140-143,150-152,161-162,2001 fast_rate mtu 9216</pre>

Leaf1A	Leaf1B
<pre> no shutdown ! mclag domain 1 source-ip 10.1.1.1 peer-ip 10.1.1.2 peer-link PortChannel100 mclag-system-mac 00:00:00:00:00:01 </pre>	<pre> no shutdown ! mclag domain 1 source-ip 10.1.1.2 peer-ip 10.1.1.1 peer-link PortChannel100 mclag-system-mac 00:00:00:00:00:01 </pre>

Configure port channels

About this task

For each port channel connected to a PowerFlex node, perform the following steps:

Steps

1. Create an LACP port channel.
2. Provide a description.
3. Assign the VLANs required by the node type to be connected to the port channel as shown in [VLAN membership by node type](#).
4. Enable LACP individual.
LACP individual: If LACPDUs are not received on an individual member port that is in link-up state, the LACP Individual feature transitions the port to lacp individual mode.
5. Set the MTU size to 9216 bytes for best performance.
6. Enable the interface.
7. Bind the port channel to the MC-LAG domain ID created in [Configure the MC-LAG domain](#).
8. Enable link state tracking for downstream links.
This setting helps to minimize switch failover time.
9. Enable spanning tree BPDU guard.
10. Set the port type to edge.
11. Enable spanning tree root guard.

Leaf1A	Leaf 1B
<pre> interface PortChannel 1 description "Bond0 to SO node" switchport trunk allowed Vlan 150-151,161 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 2 description "Bond0 to SO node" switchport trunk allowed Vlan 150-151,161 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard </pre>	<pre> interface PortChannel 1 description "Bond0 to SO node" switchport trunk allowed Vlan 150-151,161 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 2 description "Bond0 to SO node" switchport trunk allowed Vlan 150-151,161 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard </pre>

Leaf1A	Leaf 1B
<pre> spanning-tree port type edge spanning-tree guard root ! interface PortChannel 3 description "Bond0 to CO node" switchport trunk allowed Vlan 105-106 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 4 description "Bond0 to CO node" switchport trunk allowed Vlan 105-106 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 5 description "Bond1 to SO node" switchport trunk allowed Vlan 152,162 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 6 description "Bond1 to SO node" switchport trunk allowed Vlan 152,162 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 7 description "Bond1 to CO node" switchport trunk allowed Vlan 151-152 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 8 description "Bond1 to CO node" switchport trunk allowed Vlan 151-152 lacp individual mtu 9216 no shutdown mclag 1 </pre>	<pre> spanning-tree port type edge spanning-tree guard root ! interface PortChannel 3 description "Bond0 to CO node" switchport trunk allowed Vlan 105-106 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 4 description "Bond0 to CO node" switchport trunk allowed Vlan 105-106 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 5 description "Bond1 to SO node" switchport trunk allowed Vlan 152,162 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 6 description "Bond1 to SO node" switchport trunk allowed Vlan 152,162 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 7 description "Bond1 to CO node" switchport trunk allowed Vlan 151-152 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 8 description "Bond1 to CO node" switchport trunk allowed Vlan 151-152 lacp individual mtu 9216 no shutdown mclag 1 </pre>

Leaf1A	Leaf 1B
<pre> link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 9 description "FE to PFMC node" switchport trunk allowed Vlan 105,140,150 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 10 description "FE to PFMC node" switchport trunk allowed Vlan 105,140,150 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 11 description "FE to PFMC node" switchport trunk allowed Vlan 105,140,150 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 13 description "BE to PFMC node" switchport trunk allowed Vlan 103,141-143 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 14 description "BE to PFMC node" switchport trunk allowed Vlan 103,141-143 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! </pre>	<pre> link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 9 description "FE to PFMC node" switchport trunk allowed Vlan 105,140,150 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 10 description "FE to PFMC node" switchport trunk allowed Vlan 105,140,150 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 11 description "FE to PFMC node" switchport trunk allowed Vlan 105,140,150 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 13 description "BE to PFMC node" switchport trunk allowed Vlan 103,141-143 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! interface PortChannel 14 description "BE to PFMC node" switchport trunk allowed Vlan 103,141-143 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root ! </pre>

Leaf1A	Leaf 1B
<pre> interface PortChannel 15 description "BE to PFMC node" switchport trunk allowed Vlan 103,141-143 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root </pre>	<pre> interface PortChannel 15 description "BE to PFMC node" switchport trunk allowed Vlan 103,141-143 lacp individual mtu 9216 no shutdown mclag 1 link state track mclag downstream spanning-tree bpduguard spanning-tree port type edge spanning-tree guard root </pre>

Configure management and Ethernet interfaces

About this task

Perform the following steps to configure the management and Ethernet interfaces:

Steps

1. Configure the dedicated management port.
This example uses **interface Management 0**.
2. Configure an IP address on the OOB management network unique to the switch.
Specify the gateway address if routing is used on the OOB management network.

Leaf1A	Leaf1B
<pre> interface Management 0 description "OOB mgmt port" ip address 100.67.124.189/24 gwaddr 100.67.124.254 </pre>	<pre> interface Management 0 description "OOB mgmt port" ip address 100.67.124.190/24 gwaddr 100.67.124.254 </pre>

For each Ethernet interface connected to a PowerFlex node:

3. Provide a description.
4. Set the MTU size to 9216 bytes for best performance.
5. Bind the interface to the applicable port channel.
6. Enable the interface.

Leaf1A	Leaf1B
<pre> interface Eth1/1 description "To SO node NIC0:1" mtu 9216 channel-group 1 no shutdown ! interface Eth1/2 description "To SO node NIC0:1" mtu 9216 channel-group 2 no shutdown ! interface Eth1/3 description "To CO node NIC0:1" mtu 9216 </pre>	<pre> interface Eth1/1 description "To SO node NIC3:1" mtu 9216 channel-group 1 no shutdown ! interface Eth1/2 description "To SO node NIC3:1" mtu 9216 channel-group 2 no shutdown ! interface Eth1/3 description "To CO node NIC3:1" mtu 9216 </pre>

Leaf1A	Leaf1B
<pre> channel-group 3 no shutdown ! interface Eth1/4 description "To CO node NIC0:1" mtu 9216 channel-group 4 no shutdown ! interface Eth1/5 description "To SO node NIC3:2" mtu 9216 channel-group 5 no shutdown ! interface Eth1/6 description "To SO node NIC3:2" mtu 9216 channel-group 6 no shutdown ! interface Eth1/7 description "To CO node NIC3:2" mtu 9216 channel-group 7 no shutdown ! interface Eth1/8 description "To CO node NIC3:2" mtu 9216 channel-group 8 no shutdown ! interface Eth1/9 description "To PFMC node NIC0:1" mtu 9216 channel-group 9 no shutdown ! interface Eth1/10 description "To PFMC node NIC0:1" mtu 9216 channel-group 10 no shutdown ! interface Eth1/11 description "To PFMC node NIC0:1" mtu 9216 channel-group 11 no shutdown ! interface Eth1/13 description "To PFMC node NIC2:2" mtu 9216 channel-group 13 no shutdown ! interface Eth1/14 description "To PFMC node NIC2:2" mtu 9216 channel-group 14 no shutdown ! interface Eth1/15 description "To PFMC node NIC2:2" mtu 9216 channel-group 15 no shutdown </pre>	<pre> channel-group 3 no shutdown ! interface Eth1/4 description "To CO node NIC3:1" mtu 9216 channel-group 4 no shutdown ! interface Eth1/5 description "To SO node NIC0:2" mtu 9216 channel-group 5 no shutdown ! interface Eth1/6 description "To SO node NIC0:2" mtu 9216 channel-group 6 no shutdown ! interface Eth1/7 description "To CO node NIC0:2" mtu 9216 channel-group 7 no shutdown ! interface Eth1/8 description "To CO node NIC0:2" mtu 9216 channel-group 8 no shutdown ! interface Eth1/9 description "To PFMC node NIC2:1" mtu 9216 channel-group 9 no shutdown ! interface Eth1/10 description "To PFMC node NIC2:1" mtu 9216 channel-group 10 no shutdown ! interface Eth1/11 description "To PFMC node NIC2:1" mtu 9216 channel-group 11 no shutdown ! interface Eth1/13 description "To PFMC node NIC0:2" mtu 9216 channel-group 13 no shutdown ! interface Eth1/14 description "To PFMC node NIC0:2" mtu 9216 channel-group 14 no shutdown ! interface Eth1/15 description "To PFMC node NIC0:2" mtu 9216 channel-group 15 no shutdown </pre>

For each Ethernet interface that is connected to the peer leaf switch:

7. Provide a description.
8. Set the MTU size to 9216 bytes for best performance.
9. Bind the interface to the MC-LAG port channel.
This example uses **channel-group 100**.
10. Enable the interface.

Leaf1A	Leaf1B
<pre> interface Eth1/49 description "Peer link to Leaf 1B" mtu 9216 channel-group 100 no shutdown ! interface Eth1/50 description "Peer link to Leaf 1B" mtu 9216 channel-group 100 no shutdown ! interface Eth1/51 description "Peer link to Leaf 1B" mtu 9216 channel-group 100 no shutdown ! interface Eth1/52 description "Peer link to Leaf 1B" mtu 9216 channel-group 100 no shutdown </pre>	<pre> interface Eth1/49 description "Peer link to Leaf 1A" mtu 9216 channel-group 100 no shutdown ! interface Eth1/50 description "Peer link to Leaf 1A" mtu 9216 channel-group 100 no shutdown ! interface Eth1/51 description "Peer link to Leaf 1A" mtu 9216 channel-group 100 no shutdown ! interface Eth1/52 description "Peer link to Leaf 1A" mtu 9216 channel-group 100 no shutdown </pre>

 **NOTE:** If spines are not used, then Steps 11 through 15 are not required for a single rack deployment.

For each Ethernet interface connected to a spine switch:

11. Provide a description.
12. Set the MTU size to 9216 bytes for best performance.
13. Enable the interface.
14. Enable IPv6 to support unnumbered BGP point-to-point links with link-local addressing.
15. Enable link state tracking for upstream links to the spines.
This setting helps to minimize switch failover time.

Leaf1A	Leaf1B
<pre> interface Eth1/53 description "To Spine1" mtu 9216 no shutdown ipv6 enable link state track mclag upstream ! interface Eth1/54 description "To Spine2" mtu 9216 no shutdown ipv6 enable link state track mclag upstream ! interface Eth1/55 description "To Spine3" mtu 9216 </pre>	<pre> interface Eth1/53 description "To Spine1" mtu 9216 no shutdown ipv6 enable link state track mclag upstream ! interface Eth1/54 description "To Spine2" mtu 9216 no shutdown ipv6 enable link state track mclag upstream ! interface Eth1/55 description "To Spine3" mtu 9216 </pre>

Leaf1A	Leaf1B
<pre> no shutdown ipv6 enable link state track mclag upstream ! interface Eth1/56 description "To Spine4" mtu 9216 no shutdown ipv6 enable link state track mclag upstream </pre>	<pre> no shutdown ipv6 enable link state track mclag upstream ! interface Eth1/56 description "To Spine4" mtu 9216 no shutdown ipv6 enable link state track mclag upstream </pre>

On border leaf switches, configure interfaces that are connected to the external switches as needed for your environment. This configuration is used in this deployment example for access to DNS and NTP services, as shown in [PowerFlex leaf-spine production topology](#). Each interface connected to an external switch is configured in this example as follows:

16. Provide a description.
17. Set the MTU size to 9216 bytes (optional).
18. Enable the interface.
19. Bind the interface to the tenant VRF.
20. Enable IPv6 to support unnumbered BGP point-to-point links.

Leaf1A	Leaf1B
<pre> interface Eth1/47 description "To ExternalA" mtu 9216 no shutdown ip vrf forwarding VrfTenant1 ipv6 enable ! interface Eth1/48 description "To ExternalB" mtu 9216 no shutdown ip vrf forwarding VrfTenant1 ipv6 enable </pre>	<pre> interface Eth1/47 description "To ExternalA" mtu 9216 no shutdown ip vrf forwarding VrfTenant1 ipv6 enable ! interface Eth1/48 description "To ExternalB" mtu 9216 no shutdown ip vrf forwarding VrfTenant1 ipv6 enable </pre>

Configure unnumbered BGP for the nondefault VRF

About this task

Unnumbered BGP interfaces advertise IPv4 routes with an IPv6 next hop. After IPv6 is enabled on an interface that is connected to a BGP neighbor, an IPv6 link-local address is automatically created. BGP uses the link-local address to set up a BGP session with the neighbor.

Perform the following steps to configure unnumbered BGP for the nondefault (tenant) VRF:

Steps

1. Specify the router AS number and enter BGP configuration mode for the VRF tenant by running the `router bgp AS_number vrf vrf_name` command.
2. For the router ID, specify the IP address that is configured on interface Loopback 0.
3. Enter the `address-family ipv4 unicast` command.
 - a. Specify the `redistribute connected` command to redistribute IPv4 routes from physically connected interfaces.
 - b. Configure the maximum number of eBGP routes that can be used to forward packets over multiple paths to a neighbor.
4. Enter the `address-family l2vpn evpn` command.
 - a. Advertise IPv4 routes.

- b. Enable duplicate address detection.

NOTE: The following steps are for connection to the external network. In the configuration example used in this guide, the uplinks are L3 with unnumbered BGP.

5. Create a peer group for the external switch connections.

It is named LEGACY in this example.

- a. Configure a remote autonomous system to exchange routes through eBGP.
The `remote-as external` command is required for unnumbered BGP.
- b. Configure the time (in seconds) between sending keepalive messages to a BGP neighbor, and the wait time to receive a keepalive message before considering a BGP peer to be dead.
- c. Configure the route advertisement interval between sending BGP route updates to neighbors.
- d. Enable bidirectional forwarding detection (BFD).
BFD rapidly detects communication failures between two adjacent routers.
- e. The `capability extended-nexthop` command is required for unnumbered BGP.
It enables BGP to negotiate the extended next hop capability with a peer.
- f. Enter the `address-family ipv4 unicast` command, and then `activate`.
- g. Put the interfaces that are connected to the external switches in the peer group.

Leaf1A	Leaf1B
<pre> router bgp 65101 vrf VrfTenant1 router-id 10.1.1.1 ! address-family ipv4 unicast redistribute connected maximum-paths 2 ! address-family l2vpn evpn advertise ipv4 unicast dup-addr-detection peer-group LEGACY remote-as external timers 3 9 advertisement-interval 5 bfd capability extended-nexthop ! address-family ipv4 unicast activate ! neighbor interface Eth1/47 peer-group LEGACY ! neighbor interface Eth1/48 peer-group LEGACY </pre>	<pre> router bgp 65101 vrf VrfTenant1 router-id 10.1.1.2 ! address-family ipv4 unicast redistribute connected maximum-paths 2 ! address-family l2vpn evpn advertise ipv4 unicast dup-addr-detection peer-group LEGACY remote-as external timers 3 9 advertisement-interval 5 bfd capability extended-nexthop ! address-family ipv4 unicast activate ! neighbor interface Eth1/47 peer-group LEGACY ! neighbor interface Eth1/48 peer-group LEGACY </pre>

Configure unnumbered BGP for the default VRF

About this task

NOTE: If spines are not used, then these steps are not required for a single rack deployment.

Perform the following steps to configure unnumbered BGP for the default VRF:

Steps

1. Configure the BGP ASN with the `router bgp` command.
Use the same AS number that was used in [Configure unnumbered BGP for the nondefault VRF](#).

2. For the router ID, specify the IP address that is configured on interface Loopback 0.
3. Configure a maximum multi exit discriminator (MED) value to send on startup.
This setting helps to minimize switch failover time.
4. Enable ECMP with the `bestpath as-path multipath-relax` command.
5. Enter the `address-family ipv4 unicast` command.
 - a. Specify the `redistribute connected` command to redistribute IPv4 routes from physically connected interfaces.
 - b. Configure the maximum number of eBGP routes that can be used to forward packets over multiple paths to a neighbor.
6. Enter the `address-family l2vpn evpn` command.
 - a. Advertise all VNIs.
 - b. Advertise the primary IP (the router ID) of the switch.
 - c. Specify the router ID IP address of the MC-LAG peer switch.
 - d. Disable duplicate address detection.
7. Create a peer group for the spine connections.
8. Configure a remote autonomous system to exchange routes through eBGP.
The `remote-as external` command is required for unnumbered BGP.
9. Configure the time (in seconds) between sending keepalive messages to a BGP neighbor, and the wait time to receive a keepalive message before considering a BGP peer to be dead.
10. Configure the route advertisement interval between sending BGP route updates to neighbors.
11. Enable BFD.
BFD rapidly detects communication failures between two adjacent routers.
12. The `capability extended-nexthop` command is required for unnumbered BGP.
It enables BGP to negotiate the extended next hop capability with a peer.
13. Enter the `address-family ipv4 unicast` command and activate.
 - a. Set `allowas-in` to accept routes with the local AS number in updates.
14. Enter the `address-family l2vpn evpn` command and activate.
 - a. Set `allowas-in` to accept routes with the local AS number in updates.
15. Put the interfaces that are connected to the spines in the spine peer group.

Leaf1A	Leaf1B
<pre> router bgp 65101 router-id 10.1.1.1 max-med on-startup 420 bestpath as-path multipath-relax ! address-family ipv4 unicast redistribute connected maximum-paths 2 ! address-family l2vpn evpn advertise-all-vni advertise-pip ip 10.1.1.1 peer-ip 10.1.1.2 no dup-addr-detection ! peer-group SPINE remote-as external timers 3 9 advertisement-interval 5 bfd capability extended-nexthop ! address-family ipv4 unicast activate allowas-in 2 ! address-family l2vpn evpn activate allowas-in 2 </pre>	<pre> router bgp 65101 router-id 10.1.1.2 max-med on-startup 420 bestpath as-path multipath-relax ! address-family ipv4 unicast redistribute connected maximum-paths 2 ! address-family l2vpn evpn advertise-all-vni advertise-pip ip 10.1.1.2 peer-ip 10.1.1.1 no dup-addr-detection ! peer-group SPINE remote-as external timers 3 9 advertisement-interval 5 bfd capability extended-nexthop ! address-family ipv4 unicast activate allowas-in 2 ! address-family l2vpn evpn activate allowas-in 2 </pre>

Leaf1A	Leaf1B
<pre> ! neighbor interface Eth1/53 peer-group SPINE ! neighbor interface Eth1/54 peer-group SPINE ! neighbor interface Eth1/55 peer-group SPINE ! neighbor interface Eth1/56 peer-group SPINE </pre>	<pre> ! neighbor interface Eth1/53 peer-group SPINE ! neighbor interface Eth1/54 peer-group SPINE ! neighbor interface Eth1/55 peer-group SPINE ! neighbor interface Eth1/56 peer-group SPINE </pre>

Configure VXLAN

About this task

 **NOTE:** If spines are not used, then these steps are not required for a single rack deployment.

Perform the following steps to configure VXLAN:

Steps

1. Create a VXLAN interface.
2. Assign the same VTEP source IP address to each peer.
This IP address is the VTEP IP address that is configured on interface Loopback 1 earlier.
3. Assign a unique primary IP address to optimize routing for orphan ports and active-standby hosts.
This IP address can be the same IP address as the router ID.
4. Map a VNI to each VLAN that is stretched across the fabric.
Valid VNI numbers are 1 to 16777215.
5. Create a Layer 3 VNI by mapping the tenant VNI to the tenant VRF VLAN (2001 in this example).
6. Exit configuration mode and save the configuration with the `end` and `write memory` commands.

Leaf1A	Leaf1B
<pre> interface vxlan vtep1 source-ip 10.0.0.1 primary-ip 10.1.1.1 map vni 10103 vlan 103 map vni 10105 vlan 105 map vni 10106 vlan 106 map vni 10150 vlan 150 map vni 10151 vlan 151 map vni 10152 vlan 152 map vni 10161 vlan 161 map vni 10162 vlan 162 map vni 102001 vlan 2001 map vni 102001 vrf VrfTenant1 ! end write memory </pre>	<pre> interface vxlan vtep1 source-ip 10.0.0.1 primary-ip 10.1.1.2 map vni 10103 vlan 103 map vni 10105 vlan 105 map vni 10106 vlan 106 map vni 10150 vlan 150 map vni 10151 vlan 151 map vni 10152 vlan 152 map vni 10161 vlan 161 map vni 10162 vlan 162 map vni 102001 vlan 2001 map vni 102001 vrf VrfTenant1 ! end write memory </pre>

Spine switch configuration

NOTE: If spines are not used, then these steps are not required for a single rack deployment.

This section details the configuration of the Spine1 and Spine2 switches for a multirack deployment. Configuration details for the remaining spine switches in the example topology are similar, but are not shown in this chapter. Full configuration files for all switches in this deployment example are provided as attachments to this guide.

General settings

About this task

Perform these steps to configure general settings on Spine1 and Spine2:

Steps

- 1. Enter the SONiC Management Framework CLI.
- 2. Enter configuration mode.
- 3. Specify a hostname.
- 4. Change the interface naming mode to standard.
- 5. Exit back to the Linux shell to activate the changes.

Spine1	Spine2
<pre>sonic-cli configure terminal hostname Spine1 interface-naming standard end exit</pre>	<pre>sonic-cli configure terminal hostname Spine2 interface-naming standard end exit</pre>

- 6. Enter the SONiC Management Framework CLI.
- 7. Enter configuration mode.
- 8. Set the switch spanning tree mode to rapid per-VLAN spanning tree (RPVST).
- 9. Create the management VRF as a best practice.
Doing so provides an isolated routing table for the management interface. The management VRF must be named **mgmt** as shown.
- 10. Put the SSH server in the management VRF.

Spine1	Spine2
<pre>sonic-cli configure terminal spanning-tree mode rapid-pvst ip vrf mgmt ssh-server vrf mgmt</pre>	<pre>sonic-cli configure terminal spanning-tree mode rapid-pvst ip vrf mgmt ssh-server vrf mgmt</pre>

Configure the loopback interface

About this task

Perform the following steps to configure the loopback interface on Spine1 and Spine2:

Step

1. Configure a loopback interface for use as the router ID.
2. Provide a description.
3. Specify a unique IP address for each switch.

Spine1	Spine2
<pre>interface Loopback 0 description Router-ID ip address 10.2.2.1/32</pre>	<pre>interface Loopback 0 description Router-ID ip address 10.2.2.2/32</pre>

Configure management and Ethernet interfaces

About this task

Perform the following steps to configure the management and Ethernet interfaces:

Steps

1. Configure the dedicated management port **interface Management 0**.
2. Provide a description.
3. Configure an IP address and mask on the OOB management network unique to the switch.
Specify the gateway address if routing is used on the OOB management network.

Spine1	Spine2
<pre>interface Management 0 description "OOB mgmt port" ip address 100.67.124.185/24 gwaddr 100.67.124.254</pre>	<pre>interface Management 0 description "OOB mgmt port" ip address 100.67.124.186/24 gwaddr 100.67.124.254</pre>

For each Ethernet interface connected to a leaf switch:

4. Provide a description.
5. Set the MTU size to 9216 bytes for best performance.
6. Enable the interface.
7. Enable IPv6 to support unnumbered BGP point-to-point links.

Spine1	Spine2
<pre>interface Eth1/1 description "To Leaf 1A port 1/53" mtu 9216 no shutdown ipv6 enable ! interface Eth1/2 description "To Leaf 1B port 1/53" mtu 9216 no shutdown ipv6 enable ! interface Eth1/3 description "To Leaf 2A port 1/53" mtu 9216 no shutdown ipv6 enable</pre>	<pre>interface Eth1/1 description "To Leaf 1A port 1/54" mtu 9216 no shutdown ipv6 enable ! interface Eth1/2 description "To Leaf 1B port 1/54" mtu 9216 no shutdown ipv6 enable ! interface Eth1/3 description "To Leaf 2A port 1/54" mtu 9216 no shutdown ipv6 enable</pre>

Spine1	Spine2
<pre> ! interface Eth1/4 description "To Leaf 2B port 1/53" mtu 9216 no shutdown ipv6 enable ! interface Eth1/5 description "To Leaf 3A port 1/53" mtu 9216 no shutdown ipv6 enable ! interface Eth1/6 description "To Leaf 3B port 1/53" mtu 9216 no shutdown ipv6 enable ! interface Eth1/7 description "To Leaf 4A port 1/53" mtu 9216 no shutdown ipv6 enable ! interface Eth1/8 description "To Leaf 4B port 1/53" mtu 9216 no shutdown ipv6 enable </pre>	<pre> ! interface Eth1/4 description "To Leaf 2B port 1/54" mtu 9216 no shutdown ipv6 enable ! interface Eth1/5 description "To Leaf 3A port 1/54" mtu 9216 no shutdown ipv6 enable ! interface Eth1/6 description "To Leaf 3B port 1/54" mtu 9216 no shutdown ipv6 enable ! interface Eth1/7 description "To Leaf 4A port 1/54" mtu 9216 no shutdown ipv6 enable ! interface Eth1/8 description "To Leaf 4B port 1/54" mtu 9216 no shutdown ipv6 enable </pre>

Configure unnumbered BGP

About this task

Perform the following steps to configure unnumbered BGP on Spine1 and Spine2:

Steps

1. Configure the BGP ASN with the `router bgp` command.
Use the same AS number on all spine switches in the leaf-spine fabric.
2. For the router ID, specify the IP address that is configured on interface Loopback 0.
3. Enable ECMP with the `bestpath as-path multipath-relax` command.
4. Enter the `address-family ipv4 unicast` command.
 - a. Specify the `redistribute connected` command to redistribute IPv4 routes from physically connected interfaces.
 - b. Configure the maximum number of eBGP routes that can be used to forward packets over multiple paths to a neighbor.
5. Enter the `address-family l2vpn evpn` command.
 - a. Advertise all VNIs.
6. Create a peer group for the leaf connections.
 - a. Configure a remote autonomous system to exchange routes through eBGP.
The `remote-as external` command is required for unnumbered BGP.
 - b. Configure the time (in seconds) between sending keepalive messages to a BGP neighbor, and the wait time to receive a keepalive message before considering a BGP peer dead.
 - c. Configure the route advertisement interval between sending BGP route updates to neighbors.
 - d. Enable BFD.
BFD rapidly detects communication failures between two adjacent routers.
 - e. The `capability extended-nexthop` command is required for unnumbered BGP.

- f. Enter the `address-family ipv4 unicast` command and activate.
- g. Enter the `address-family l2vpn evpn` command and activate.
7. Put the interfaces that are connected to the leaf switches in the leaf peer group.
8. Exit configuration mode and save the configuration with the `end` and `write memory` commands.

Spine1	Spine2
<pre> router bgp 65200 router-id 10.2.2.1 bestpath as-path multipath-relax ! address-family ipv4 unicast redistribute connected maximum-paths 2 ! address-family l2vpn evpn advertise-all-vni ! peer-group LEAF remote-as external timers 3 9 advertisement-interval 5 bfd capability extended-nexthop ! address-family ipv4 unicast activate ! address-family l2vpn evpn activate ! neighbor interface Eth1/1 peer-group LEAF ! neighbor interface Eth1/2 peer-group LEAF ! neighbor interface Eth1/3 peer-group LEAF ! neighbor interface Eth1/4 peer-group LEAF ! neighbor interface Eth1/5 peer-group LEAF ! neighbor interface Eth1/6 peer-group LEAF ! neighbor interface Eth1/7 peer-group LEAF ! neighbor interface Eth1/8 peer-group LEAF ! end write memory </pre>	<pre> router bgp 65200 router-id 10.2.2.2 bestpath as-path multipath-relax ! address-family ipv4 unicast redistribute connected maximum-paths 2 ! address-family l2vpn evpn advertise-all-vni ! peer-group LEAF remote-as external timers 3 9 advertisement-interval 5 bfd capability extended-nexthop ! address-family ipv4 unicast activate ! address-family l2vpn evpn activate ! neighbor interface Eth1/1 peer-group LEAF ! neighbor interface Eth1/2 peer-group LEAF ! neighbor interface Eth1/3 peer-group LEAF ! neighbor interface Eth1/4 peer-group LEAF ! neighbor interface Eth1/5 peer-group LEAF ! neighbor interface Eth1/6 peer-group LEAF ! neighbor interface Eth1/7 peer-group LEAF ! neighbor interface Eth1/8 peer-group LEAF ! end write memory </pre>

Verify leaf switch configuration

The following commands and outputs are from Leaf1A in a four-rack deployment. Most commands also apply to single rack deployments except where noted. Command output on other leaf switches in the leaf-spine fabric is similar. For the following command output examples, PowerFlex nodes are connected and powered on but are not deployed yet. Changes to expect in the command output after deploying PowerFlex nodes are noted where applicable. Lengthy CLI output is truncated where noted.

show interface status

Ensure that Ethernet interfaces that are connected to PowerFlex nodes and switches are up, and are at the expected speed and MTU. Port channels connected to PowerFlex nodes may show up on one leaf in the pair, but are down with the reason `lACP-fail` on the other leaf since the nodes are not yet deployed. After PowerFlex nodes are deployed, the applicable port channels should be up on both leaf switches.

```
Leaf1A# show interface status
```

Name	Description	Oper	Reason	AutoNeg	Speed	MTU
Eth1/1	To SO node NIC0:1	up	oper-up	off	25000	9216
Eth1/2	To SO node NIC0:1	up	oper-up	off	25000	9216
(output truncated)						
Eth1/47	Link to external network	up	oper-up	off	25000	9216
Eth1/48	Link to external network	up	oper-up	off	25000	9216
Eth1/49	Peer link to Leaf 1B	up	oper-up	off	100000	9216
Eth1/50	Peer link to Leaf 1B	up	oper-up	off	100000	9216
Eth1/51	Peer link to Leaf 1B	up	oper-up	off	100000	9216
Eth1/52	Peer link to Leaf 1B	up	oper-up	off	100000	9216
Eth1/53	To Spine1	up	oper-up	off	100000	9216
Eth1/54	To Spine2	up	oper-up	off	100000	9216
Eth1/55	To Spine3	up	oper-up	off	100000	9216
Eth1/56	To Spine4	up	oper-up	off	100000	9216
PortChannel1	Bond0 to SO node	down	lACP-fail	-	25000	9216
PortChannel2	Bond0 to SO node	down	lACP-fail	-	25000	9216
(output truncated)						
PortChannel100	MCLAG peer link	up	oper-up	-	400000	9216

show lldp table

The `show lldp table` command is used to verify connections are connected to the correct devices. For example, the command output on Leaf1A shows the MAC address of each PowerFlex NIC connected to interfaces Eth1/1-1/15. The MAC shown in the command output can be cross-referenced with the MAC address that is shown in the PowerFlex node iDRAC during troubleshooting if necessary. Remote switch host names and ports are shown on interfaces Eth1/47-1/56 in this example.

```
Leaf1A# show lldp table
```

LocalPort	RemoteDevice	RemotePortID	RemotePortDescr
Eth1/1		e8:eb:d3:29:88:f0	ConnectX-5,...
Eth1/2		e8:eb:d3:29:88:50	ConnectX-5,...
Eth1/3		e8:eb:d3:25:50:90	ConnectX-5,...
Eth1/4		e8:eb:d3:05:ef:64	ConnectX-5,...
Eth1/5		b8:3f:d2:b9:80:3b	ConnectX-5,...
Eth1/6		b8:3f:d2:b9:80:3f	ConnectX-5,...
Eth1/7		b8:3f:d2:b9:80:27	ConnectX-5,...
Eth1/8		b8:3f:d2:b9:7e:ab	ConnectX-5,...
Eth1/9		e8:eb:d3:05:ec:b4	ConnectX-5,...
Eth1/10		10:70:fd:d4:12:8c	ConnectX-5,...
Eth1/11		e8:eb:d3:06:31:cc	ConnectX-5,...
Eth1/13		e8:eb:d3:8a:93:eb	ConnectX-5,...
Eth1/14		e8:eb:d3:27:ea:a5	ConnectX-5,...
Eth1/15		e8:eb:d3:27:ea:ad	ConnectX-5,...
Eth1/47	ExternalA	Eth1/9	To Leaf1A port 1/47
Eth1/48	ExternalB	Eth1/9	To Leaf1A port 1/48
Eth1/49	Leaf1B	Eth1/49	Peer link to Leaf 1A
Eth1/50	Leaf1B	Eth1/50	Peer link to Leaf 1A
Eth1/51	Leaf1B	Eth1/51	Peer link to Leaf 1A
Eth1/52	Leaf1B	Eth1/52	Peer link to Leaf 1A
Eth1/53	Spine1	Eth1/1	To Leaf 1A port 1/53
Eth1/54	Spine2	Eth1/1	To Leaf 1A port 1/54
Eth1/55	Spine3	Eth1/1	To Leaf 1A port 1/56
Eth1/56	Spine4	Eth1/1	To Leaf 1A port 1/56

show mclag brief

The `show mclag brief` command shows the status of the MC-LAG domain. One switch has the role set to active and the other standby. Session Status and Peer Link Status should both be up. The Mclag System Mac must be the same on both leaf switches in the pair. It must be different from all other leaf pairs and devices in the leaf-spine fabric.

Before PowerFlex node deployment, the MC-LAG interface status for the PowerFlex port channels shows down/up or up/down. After the PowerFlex nodes are deployed, the status for the connected port channels should change to up/up.

```
Leaf1A# show mclag brief

Domain ID      : 1
Role           : active
Session Status : up
Peer Link Status : up
Source Address : 10.1.1.1
Peer Address   : 10.1.1.2
Peer Link      : PortChannel100
Keepalive Interval : 1 secs
Session Timeout : 30 secs
Delay Restore  : 300 secs
System Mac     : b0:4f:13:2b:6a:a2
Mclag System Mac : 00:00:00:00:00:01

Number of MLAG Interfaces:14
-----
MLAG Interface      Local/Remote Status
-----
PortChannel1       down/up
PortChannel2       down/up
(output truncated)
```

show vlan

This command shows all VLANs and associated interfaces on the switch. You can use this command to ensure PowerFlex node port channels are tagged in the correct VLANs. The MC-LAG port channel, **PortChannel100** in this example, should be tagged in all VLANs.

In multirack deployments using BGP EVPN VXLAN, VLANs that are stretched between racks should be indicated with the VTEP address of the other leaf pairs that use that VLAN. For example, VLANs 105, 106, 150 to 152, and 161 to 162 include `Vxlan_10.0.0.2`, `10.0.0.3`, and `10.0.0.4` in the Ports column. `10.0.0.2`, `10.0.0.3`, and `10.0.0.4` are the VTEP IP addresses of the leaf pairs in Racks 2, 3, and 4 respectively. VLANs 103 and 140 to 143 are only required for PFMC nodes and are limited to Rack 1 in this example, so no VTEP addresses are shown.

```
Leaf1A# show Vlan

Q: A - Access (Untagged), T - Tagged
NUM      Status      Q Ports      Autostate    Dynamic
103      Active      T PortChannel13  Enable      No
          T PortChannel14
          T PortChannel15
          T PortChannel100
105      Active      T PortChannel3   Enable      No
          T PortChannel4
          T PortChannel9
          T PortChannel10
          T PortChannel11
          T PortChannel100
          A Vxlan_10.0.0.2
          A Vxlan_10.0.0.3
          A Vxlan_10.0.0.4
          T PortChannel13
          T PortChannel4
          T PortChannel100
          A Vxlan_10.0.0.2
```


140	Active	A	Vxlan_10.0.0.3		No
		A	Vxlan_10.0.0.4		No
		T	PortChannel9	Enable	No
		T	PortChannel10		No
141	Active	T	PortChannel11		No
		T	PortChannel100		No
		T	PortChannel13	Enable	No
		T	PortChannel14		No
142	Active	T	PortChannel15		No
		T	PortChannel100		No
		T	PortChannel13	Enable	No
		T	PortChannel14		No
143	Active	T	PortChannel15		No
		T	PortChannel100		No
		T	PortChannel13	Enable	No
		T	PortChannel14		No
150	Active	T	PortChannel15		No
		T	PortChannel100		No
		T	PortChannel11	Enable	No
		T	PortChannel12		No
151	Active	T	PortChannel9		No
		T	PortChannel10		No
		T	PortChannel11		No
		T	PortChannel100		No
152	Active	A	Vxlan_10.0.0.2		No
		A	Vxlan_10.0.0.3		No
		A	Vxlan_10.0.0.4		No
		T	PortChannel11	Enable	No
161	Active	T	PortChannel12		No
		T	PortChannel7		No
		T	PortChannel8		No
		T	PortChannel100		No
162	Active	A	Vxlan_10.0.0.2		No
		A	Vxlan_10.0.0.3		No
		A	Vxlan_10.0.0.4		No
		T	PortChannel5	Enable	No
2001	Active	T	PortChannel6		No
		T	PortChannel7		No
		T	PortChannel8		No
		T	PortChannel100		No

show ip route

The `show ip route` command shows the routes in the default VRF. The output includes all leaf pair VTEP IP addresses and leaf and spine switch router IDs. The IPv6 addresses shown are automatically configured link-local addresses that are used as part of the unnumbered BGP feature for the leaf-spine connections.

```
Leaf1A# show ip route
```

```
Codes:  K - kernel route, C - connected, S - static, B - BGP, O - OSPF
        > - selected route, * - FIB route, q - queued route, r - rejected route
        Destination          Gateway                      Dist/Metric
-----
```

C>*	10.0.0.1/32	Direct	Loopback1	0/0
B>*	10.0.0.2/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.0.0.3/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.0.0.4/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
C>*	10.1.1.1/32	Direct	Loopback0	0/0
B>*	10.1.1.2/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.1.1.3/32	via fe80::3e2c:30ff:fe72:7402	Eth1/56	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.1.1.4/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.1.1.5/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.1.1.6/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.1.1.7/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.1.1.8/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
*		via fe80::b24f:13ff:fe9d:cd22	Eth1/53	
B>*	10.2.2.1/32	via fe80::b24f:13ff:fe9d:cd22	Eth1/53	20/0
B>*	10.2.2.2/32	via fe80::b24f:13ff:fe9d:6e22	Eth1/54	20/0
B>*	10.2.2.3/32	via fe80::8e04:baff:fea9:4642	Eth1/55	20/0
B>*	10.2.2.4/32	via fe80::3e2c:30ff:fe72:7402	Eth1/56	20/0

show ip route vrf vrf_name

The `show ip route vrf vrf_name` command shows the routes in the nondefault VRF. The output includes all PowerFlex Layer 3 networks. In this example, it also includes the Router IDs of the external switches, and the external Layer 3 network, 192.168.10.0. This route is used to external services such as DNS and NTP. The IPv6 addresses shown are automatically configured link-local addresses that are used as part of the unnumbered BGP feature for the external switch connections.

Leaf1A# show ip route vrf VrfTenant1

Codes: K - kernel route, C - connected, S - static, B - BGP, O - OSPF				
> - selected route, * - FIB route, q - queued route, r - rejected route				
	Destination	Gateway		Dist/Metric

B>*	10.3.0.1/32	via fe80::3e2c:30ff:fe84:8302	Eth1/47	20/0
B>*	10.3.0.2/32	via fe80::3e2c:30ff:fe84:8482	Eth1/48	20/0
B>*	192.168.10.0/24	via fe80::3e2c:30ff:fe84:8302	Eth1/47	20/0
*		via fe80::3e2c:30ff:fe84:8482	Eth1/48	
B>*	192.168.10.254/32	via fe80::3e2c:30ff:fe84:8482	Eth1/48	20/0
C>*	192.168.105.0/24	Direct	Vlan105	0/0
C>*	192.168.140.0/24	Direct	Vlan140	0/0
C>*	192.168.150.0/24	Direct	Vlan150	0/0
C>*	192.168.161.0/24	Direct	Vlan161	0/0
C>*	192.168.162.0/24	Direct	Vlan162	0/0

In multitrack environments, after PowerFlex nodes are deployed, the `Destination` column will also include the IP addresses of PowerFlex nodes on Layer 3 networks in other racks. The `Gateway` column will show the remote VTEP IP address that is used to reach the node. The `Interfaces` column will list the VRF tenant VLAN, 2001 in this example.

show ip vrf

Use the `show ip vrf` command to ensure that Ethernet and VLAN interfaces are in the correct VRF.

Leaf1A# show ip vrf

VRF-NAME	INTERFACES

VrfTenant1	Eth1/47
	Eth1/48

```

Vlan103
Vlan105
Vlan106
Vlan140
Vlan141
Vlan142
Vlan143
Vlan150
Vlan151
Vlan152
Vlan153
Vlan154
Vlan161
Vlan162
Vlan2001
default  Eth1/53
          Eth1/54
          Eth1/55
          Eth1/56
          Loopback0
          Loopback1
mgmt      Management0

```

show vxlan tunnel

For multirack deployments, use this command to ensure VXLAN tunnels are established between the VTEPs in all racks of the leaf-spine fabric. The source IP (SIP) is the VTEP address for this leaf pair in Rack 1, 10.0.0.1. The destination IP (DIP) addresses 10.0.0.2, 10.0.0.3, and 10.0.0.4 are the VTEP IPs for the leaf pairs in Racks 2, 3, and 4 respectively.

```
Leaf1A# show vxlan tunnel
```

Name =====	SIP ===	DIP ===	source =====	Group =====	D-VNI =====	operstatus =====
EVPN_10.0.0.2	10.0.0.1	10.0.0.2	EVPN	internal	no	oper_up
EVPN_10.0.0.3	10.0.0.1	10.0.0.3	EVPN	internal	no	oper_up
EVPN_10.0.0.4	10.0.0.1	10.0.0.4	EVPN	internal	no	oper_up

show vxlan remote vni

For multirack deployments, use the `show vxlan remote vni` command to ensure VLANs are stretched between the required racks. For example, VLAN 105 is stretched between 10.0.0.2, 10.0.0.3, and 10.0.0.4. These IP addresses are the leaf pair VTEP IP addresses for Racks 2, 3, and 4 respectively. It also provides the VNI assigned to each VLAN in the fabric.

```
Leaf1A# show vxlan remote vni
```

Vlan =====	Tunnel =====	Group =====	VNI ===
Vlan105	10.0.0.2	internal	10105
Vlan105	10.0.0.3	internal	10105
Vlan105	10.0.0.4	internal	10105
Vlan106	10.0.0.2	internal	10106
Vlan106	10.0.0.3	internal	10106
Vlan106	10.0.0.4	internal	10106
Vlan150	10.0.0.2	internal	10150
Vlan150	10.0.0.3	internal	10150
Vlan150	10.0.0.4	internal	10150
Vlan151	10.0.0.2	internal	10151
Vlan151	10.0.0.3	internal	10151
Vlan151	10.0.0.4	internal	10151
Vlan152	10.0.0.2	internal	10152
Vlan152	10.0.0.3	internal	10152
Vlan152	10.0.0.4	internal	10152
Vlan153	10.0.0.2	internal	10153
Vlan153	10.0.0.3	internal	10153
Vlan153	10.0.0.4	internal	10153

```

Vlan154      10.0.0.2      internal  10154
Vlan154      10.0.0.3      internal  10154
Vlan154      10.0.0.4      internal  10154
Vlan161      10.0.0.2      internal  10161
Vlan161      10.0.0.3      internal  10161
Vlan161      10.0.0.4      internal  10161
Vlan162      10.0.0.2      internal  10162
Vlan162      10.0.0.3      internal  10162
Vlan162      10.0.0.4      internal  10162
Total count :    27

```

Verify spine switch configuration

 **NOTE:** This section is for multirack deployments only.

The following commands and output are from Spine1 in this four-rack and four-spine deployment example. Command output on other spine switches in the leaf-spine fabric is similar.

show interface status

Use this command to ensure Ethernet interfaces that are connected to the leaf switches are up and at the expected speed and MTU.

```
Spine1# show interface status
```

Name	Description	Oper	Reason	AutoNeg	Speed	MTU
Eth1/1	To Leaf 1A port 1/53	up	oper-up	off	100000	9216
Eth1/2	To Leaf 1B port 1/53	up	oper-up	off	100000	9216
Eth1/3	To Leaf 2A port 1/53	up	oper-up	off	100000	9216
Eth1/4	To Leaf 2B port 1/53	up	oper-up	off	100000	9216
Eth1/5	To Leaf 3A port 1/53	up	oper-up	off	100000	9216
Eth1/6	To Leaf 3B port 1/53	up	oper-up	off	100000	9216
Eth1/7	To Leaf 4A port 1/53	up	oper-up	off	100000	9216
Eth1/8	To Leaf 4B port 1/53	up	oper-up	off	100000	9216

show lldp table

The show lldp table command output on Spine1 confirms interfaces Eth1/1-1/8 are connected to the eight leaf switches in the fabric.

```
Spine1# show lldp table
```

LocalPort	RemoteDevice	RemotePortID	RemotePortDescr
Eth1/1	Leaf1A	Eth1/53	To Spine1
Eth1/2	Leaf1B	Eth1/53	To Spine1
Eth1/3	Leaf2A	Eth1/53	To Spine1
Eth1/4	Leaf2B	Eth1/53	To Spine1
Eth1/5	Leaf3A	Eth1/53	To Spine1
Eth1/6	Leaf3B	Eth1/53	To Spine1
Eth1/7	Leaf4A	Eth1/53	To Spine1
Eth1/8	Leaf4B	Eth1/53	To Spine1

show ip route

The `show ip route` command shows the routes in the default VRF. The output includes all leaf pair VTEP IP addresses and router IDs. The IPv6 addresses shown are automatically configured link-local addresses that are used as part of the unnumbered BGP feature for the leaf-spine connections.

```
Spinel# show ip route
```

```
Codes:  K - kernel route, C - connected, S - static, B - BGP, O - OSPF
> - selected route, * - FIB route, q - queued route, r - rejected route
Destination      Gateway
-----
```

B>*	10.0.0.1/32	via fe80::8e47:beff:fead:2311	Eth1/2	20/0
*		via fe80::b24f:13ff:fe2b:6aa2	Eth1/1	
B>*	10.0.0.2/32	via fe80::b24f:13ff:fe2b:5ea2	Eth1/4	20/0
*		via fe80::b24f:13ff:fe2b:89a2	Eth1/3	
B>*	10.0.0.3/32	via fe80::b24f:13ff:fe37:9c2	Eth1/5	20/0
*		via fe80::b24f:13ff:fe37:fc2	Eth1/6	
B>*	10.0.0.4/32	via fe80::b24f:13ff:fe36:50c2	Eth1/7	20/0
*		via fe80::b24f:13ff:fe36:df42	Eth1/8	
B>*	10.1.1.1/32	via fe80::b24f:13ff:fe2b:6aa2	Eth1/1	20/0
B>*	10.1.1.2/32	via fe80::8e47:beff:fead:2311	Eth1/2	20/0
B>*	10.1.1.3/32	via fe80::b24f:13ff:fe2b:89a2	Eth1/3	20/0
B>*	10.1.1.4/32	via fe80::b24f:13ff:fe2b:5ea2	Eth1/4	20/0
B>*	10.1.1.5/32	via fe80::b24f:13ff:fe37:9c2	Eth1/5	20/0
B>*	10.1.1.6/32	via fe80::b24f:13ff:fe37:fc2	Eth1/6	20/0
B>*	10.1.1.7/32	via fe80::b24f:13ff:fe36:50c2	Eth1/7	20/0
B>*	10.1.1.8/32	via fe80::b24f:13ff:fe36:df42	Eth1/8	20/0
C>*	10.2.2.1/32	Direct	Loopback0	0/0

Deploy PowerFlex

Topics:

- [Deployment notes](#)

Deployment notes

- After the network has been configured and verified, PowerFlex can be deployed according to your PowerFlex Deployment and Administration documentation.
- In PowerFlex Manager, be sure to use a partial network automation template, such as **Compute-Linux-Partial Network** or **Storage-Partial Network**, as a starting point.

 **NOTE:** Using a PowerFlex template that does not include "Partial Network" in the name may result in deployment failures.

Using an access-aggregation topology with PowerFlex

Topics:

- [Introduction](#)
- [Access-aggregation topology networking requirements](#)
- [Topology differences](#)
- [Switch configuration file examples](#)

Introduction

Dell Technologies recommends that you use the leaf-spine production topology covered in this guide. A leaf-spine topology is the standard data center architecture, is designed to optimize performance, and is highly scalable. However, in smaller deployments, some customers might prefer to use an access-aggregation production topology, where the aggregation switches terminate all Layer 2 traffic from the PowerFlex cluster and connect to core switches for North-South communication. The supported access-aggregation topology for multirack PowerFlex deployments is shown in the figure below.

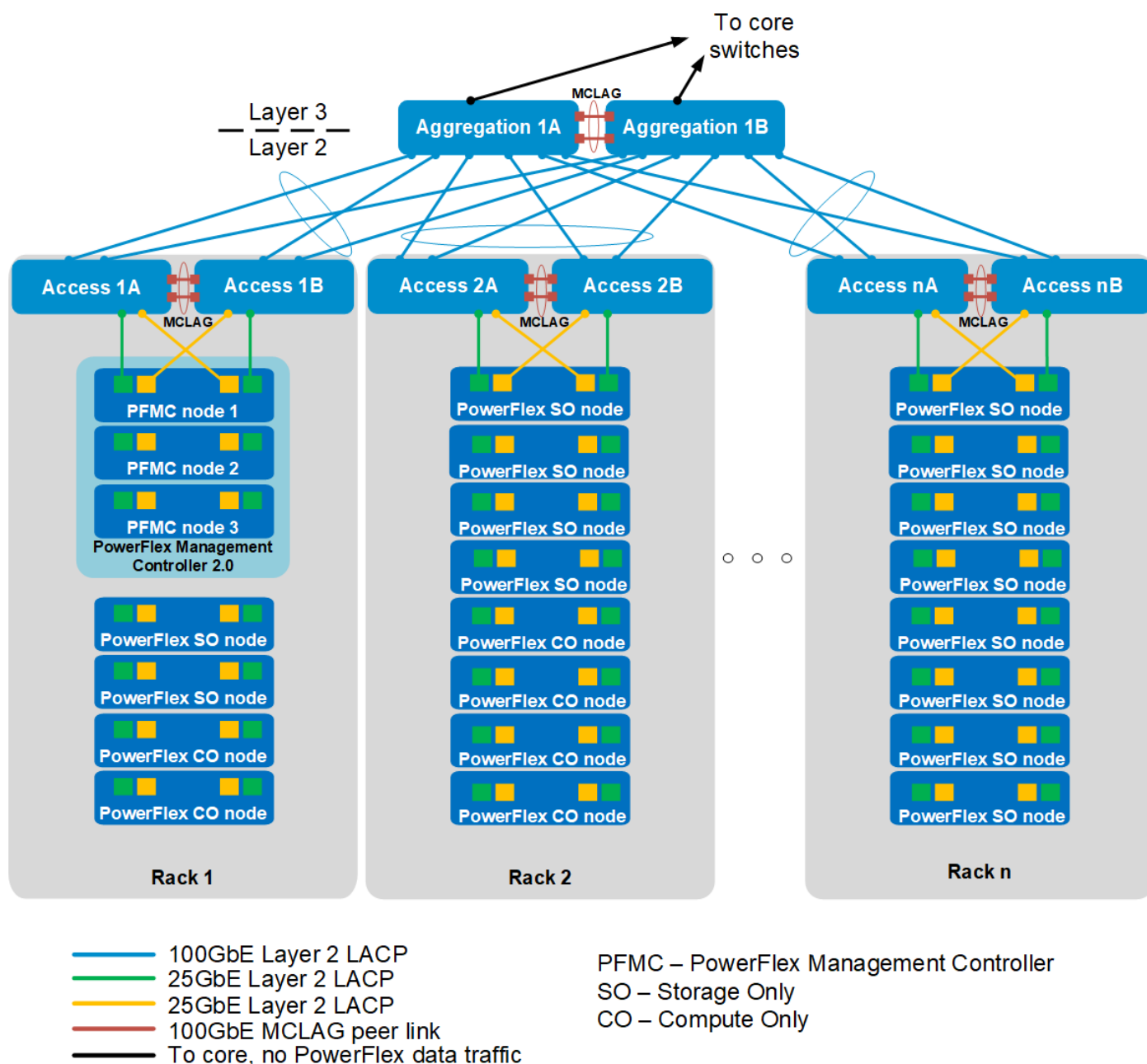


Figure 10. Access-aggregation topology for multirack PowerFlex deployments

NOTE: The OOB management topology shown in this guide is the same for access-aggregation and leaf-spine production topologies.

Access-aggregation topology networking requirements

Requirements for PowerFlex on SONiC access-aggregation topology include the following:

NOTE: "Single rack" refers to topologies with only two access switches. "Multirack" refers to topologies with four or more access switches.

- All access and aggregation switches must be running the same version of Dell Enterprise SONiC, and it must be 4.2.0 or later. An Enterprise Standard or Enterprise Premium bundle is required.

- Single rack deployments require two supported Dell PowerSwitch access switches. S5212F-ON, S5224F-ON, and S5248F-ON are supported as access switches.
- Multirack deployments require two supported Dell PowerSwitch access switches per rack, plus two supported Dell PowerSwitch aggregation switches. S5232F-ON is supported as an aggregation switch.
- Access and aggregation switches must be deployed in pairs and be configured for MC-LAG. Each switch in an MC-LAG pair must be the same model.
- In multirack deployments, all access switches connected to PowerFlex nodes must be connected to the same pair of aggregation switches.
- If aggregation switches have uplinks to core switches, PowerFlex data traffic must not pass through the core switches.
- Each access-to-aggregation switch connection must be at least 100 GbE, Layer 2, and be configured for LACP. Provide enough connections to ensure the oversubscription ratio at the access switches does not exceed 2:1.
- MC-LAG peer links between access and aggregation switch pairs require at least two connections for fault tolerance. Each access switch peer link can be 25 GbE SFP28, 100 GbE QSFP28, or 200 GbE QSFP28-DD (available on S5248F-ON). If 25 GbE ports are used, Dell Technologies recommends using at least four connections for sufficient bandwidth if a switch is taken offline. Each aggregation switch peer link must be at least 100 GbE.
- Each PowerFlex node requires four 25 GbE SFP28 connections to the access switches. LACP is recommended on these connections, but it is not required.
- One 1 GbE (1000BASE-T) switch for OOB management connections is required. Dell Technologies recommends using one PowerSwitch S3248T-ON per rack.
- One 1 GbE (1000BASE-T) connection is required for each switch OOB management port.
- One 1 GbE (1000BASE-T) connection is required for each PowerFlex node iDRAC port.
- One 10 GbE connection is required for each PFMC node to access the OOB management network in addition to the PFMC node's 1 GbE iDRAC port.
- One DNS server and one NTP server that are reachable from the PowerFlex Node Management network are required.

Topology differences

The following table shows the key differences between an access-aggregation topology and a leaf-spine topology for PowerFlex on Dell Enterprise SONiC.

Table 7. Topology differences

Item	Leaf-spine topology (preferred)	Access-aggregation topology
Maximum number of access or leaf switches	32 (16 pairs)	14 (7 pairs)
Maximum number of aggregation or spine switches	6	2 (1 pair)
Maximum number of PowerFlex nodes per access or leaf pair	22	22
Maximum number of PowerFlex nodes in the topology	352	154
Layer 2 / Layer 3 boundary	Leaf layer	Aggregation layer
BGP EVPN protocol	Used	Not used

Switch configuration file examples

Example switch configuration files for the access-aggregation topology are attached to this guide. The file names begin with *x_access* and *x_aggregation* as applicable. To access the files, see the [Attachments](#) section of this guide.

Validated components

Topics:

- [Introduction](#)
- [Dell PowerSwitch systems](#)
- [PowerFlex nodes](#)
- [PowerFlex Management Controller](#)

Introduction

The following tables include the hardware, software, and firmware that were used to validate the examples in this guide. Other supported hardware, software, and firmware can be used.

Dell PowerSwitch systems

Table 8. Dell PowerSwitch models and operating systems used to validate the leaf-spine example

Quantity	Item	Role	SONiC version
8	Dell PowerSwitch S5248F-ON	Leaf switch	4.1.1 Enterprise Premium
4	Dell PowerSwitch S5232F-ON	Spine switch	4.1.1 Enterprise Premium
2	Dell PowerSwitch S5212F-ON	External switch	4.1.1 Enterprise Premium
4	Dell PowerSwitch S3248T-ON	OOB management switch	4.1.1 Enterprise Premium

Table 9. Dell PowerSwitch models and operating systems used to validate the access-aggregation example

Quantity	Item	Role	SONiC version
8	Dell PowerSwitch S5248F-ON	Access switch	4.2.0 Enterprise Standard
2	Dell PowerSwitch S5232F-ON	Aggregation switch	4.2.0 Enterprise Standard
4	Dell PowerSwitch S3248T-ON	OOB management switch	4.2.0 Enterprise Standard

PowerFlex nodes

Table 10. PowerFlex R650 S (Storage) node configuration

Quantity per node	Item	Firmware version
2	Intel Xeon Gold 6326 CPU @ 2.90 GHz, 16 cores	NA
8	32 GB DDR4 DIMM (256 GB total)	NA
5	3576.98 GB SATA SSD (HBA355i front controller)	J404

Table 10. PowerFlex R650 S (Storage) node configuration (continued)

Quantity per node	Item	Firmware version
2	447.13 GB SATA SSD (BOSS-S2 controller)	DL74
1	Dell HBA355i front storage controller	17.15.08.00
1	Boot Optimized Storage Solution (BOSS-S2) controller	2.5.13.4008
1	Broadcom NetXtreme Gigabit Ethernet dual port NIC	22.31.6
2	Mellanox ConnectX-5 EN 25 GbE dual-port SFP28 adapter	16.32.20.04
-	BIOS	1.8.2
-	CPLD	1.0.9
-	iDRAC with Lifecycle Controller	6.00.30.202
-	Backplane	3.67

Table 11. PowerFlex R650 C (Compute) node configuration

Quantity per node	Item	Firmware version
2	Intel Xeon Gold 6326 CPU @ 2.90 GHz, 16 cores	NA
8	32 GB DDR4 DIMM (256 GB total)	NA
2	447.13 GB SATA SSD	DL74
1	Boot Optimized Storage Solution (BOSS-S2) controller	2.5.13.4008
1	Broadcom NetXtreme Gigabit Ethernet dual port NIC	22.31.6
2	Mellanox ConnectX-5 EN 25 GbE dual-port SFP28 adapter	16.32.20.04
-	BIOS	1.8.2
-	CPLD	1.0.9
-	iDRAC with Lifecycle Controller	6.00.30.202

PowerFlex Management Controller

Three PowerFlex R650 S (Storage) nodes and the software in the table below were used to build the PowerFlex Management Controller 2.0 cluster.

Table 12. PowerFlex Management Controller software

Item	Version
VMware ESXi	7.0.3 build 19482537
VMware vCenter Server Appliance	7.0.3 build 19480866
PowerFlex Manager	4.0.1 build 249
Intelligent Catalog	40.371.00

Resources

Topics:

- [Dell Networking resources](#)
- [PowerFlex guides](#)
- [Fabric Design Center](#)
- [Technical support](#)

Dell Networking resources

- [Dell Networking Solutions Info Hub](#)
- [Dell Enterprise SONiC Distribution \(account required\)](#)
- [Dell PowerSwitch N3200-ON Series Documentation](#)
- [Dell PowerSwitch S5212F-ON Documentation](#)
- [Dell PowerSwitch S5248F-ON Documentation](#)
- [Dell PowerSwitch S5232F-ON Documentation](#)

PowerFlex guides

Contact your Dell account representative for access to the following documents:

- *Dell PowerFlex Appliance with PowerFlex 4.x Network Planning Guide*
- *Dell PowerFlex Appliance with PowerFlex 4.x Deployment Guide*

Fabric Design Center

The Dell Fabric Design Center (FDC) is a cloud-based application that automates the planning, design, and deployment of network fabrics that power Dell compute, storage, and hyperconverged infrastructure solutions. The FDC is ideal for turnkey solutions and automation based on validated deployment guides.

FDC allows design customization and flexibility to go beyond validated deployment guides. For additional information, go to the Dell Fabric Design Center at <https://fdc.dell.com/>.

Technical support

For technical support, visit the [Dell Support site](#).